

$\textcircled{n^2}$



length $\rightarrow$ $\textcircled{n}$ ✓

$H - 1$ max
 $\textcircled{\times}$

$0 \dots n$
 $\underbrace{\hspace{1cm}}$
 $\hookrightarrow \underline{n+1} \rightarrow (n)$

$arr \rightarrow [0, 5, 4, 6, 3] \rightarrow \underline{n=6}$

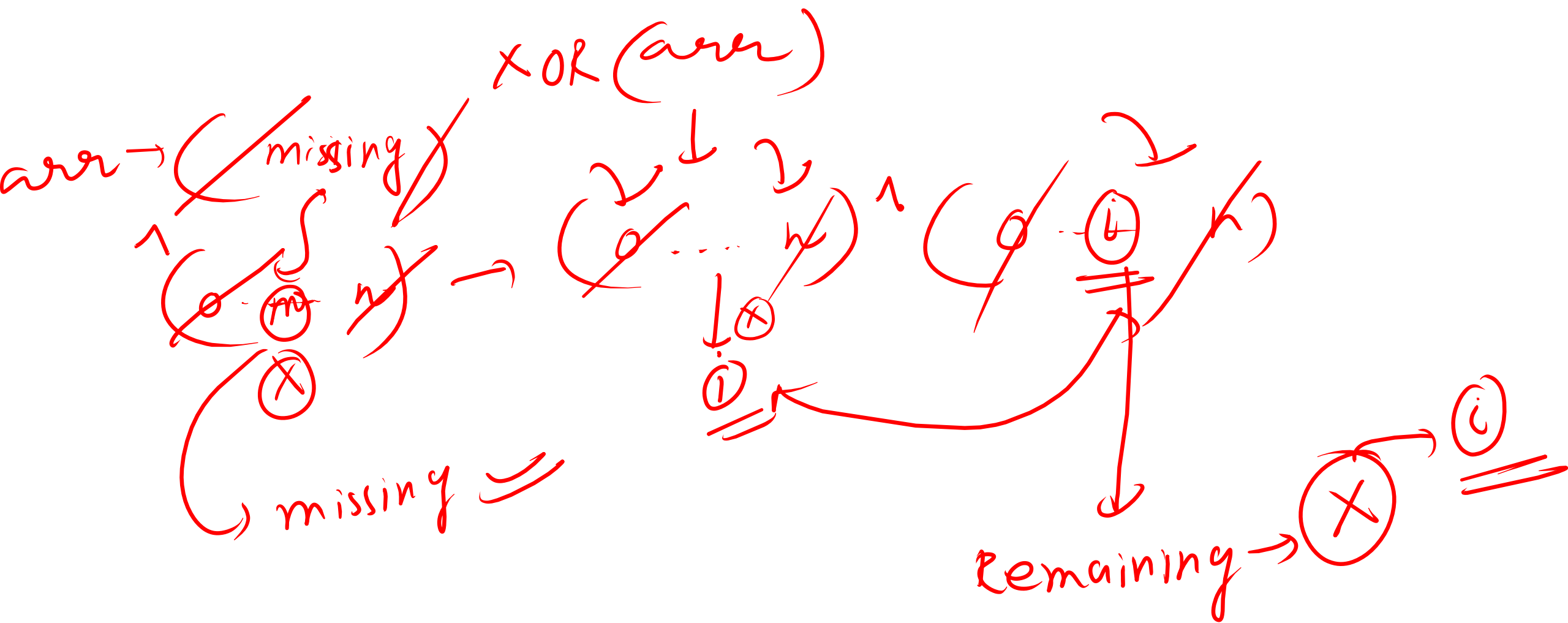
$0 \dots n$
 $\textcircled{0}$
 $\downarrow$
missing

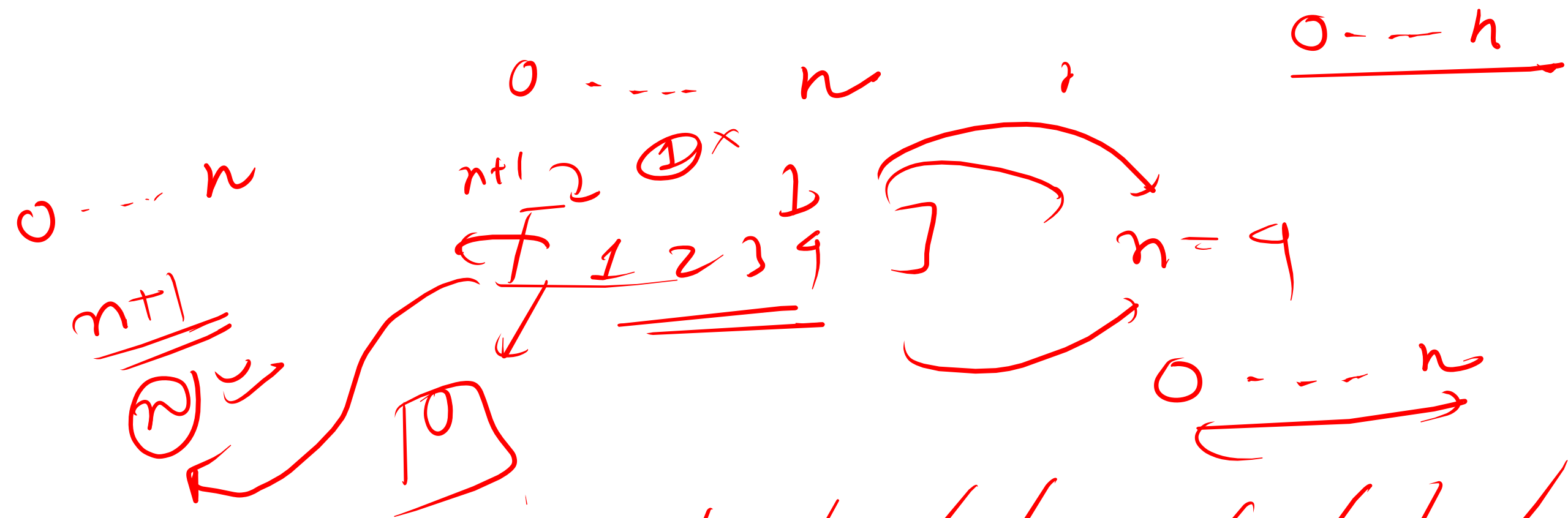
$\underline{\underline{x}} = (0 \wedge 0 \wedge 1 \wedge 1 \wedge 5 \wedge 4 \wedge 6 \wedge 3)$

$1 \wedge 0 \wedge 1 \wedge 1 \wedge \textcircled{2} \wedge 1 \wedge 2 \wedge 4 \wedge 5 \wedge 6$

$0 \dots n$
 $\longleftarrow \hspace{1cm} \longrightarrow$

$x = (2)$
 $x \wedge (0 \dots n)$
 $x \wedge arr$



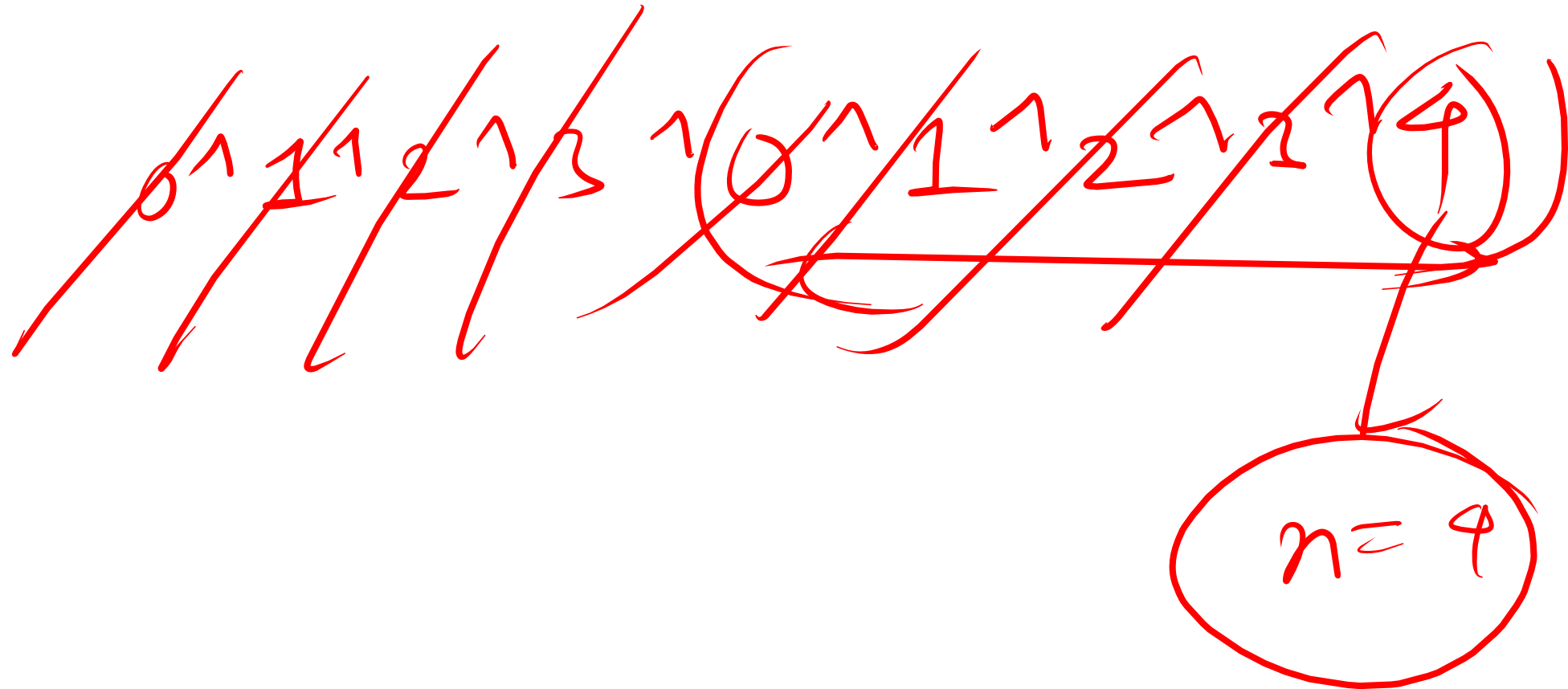


$x = 1, 2, 3, 4, 1, 0, 1, 2, 3, 4$

↓

0

[0 1 2 3] n=4



Powers of 4

$$\underline{\underline{4^3 = 64}}$$

$$\underline{\underline{10000000}}$$

$$8$$

$$\underline{\underline{4^0 \rightarrow 1 \checkmark}}$$

$$0 \dots \underline{\underline{1}}_0^2$$

$$\underline{\underline{4^1 \rightarrow 4 \rightarrow}}$$

$$0 \dots \underline{\underline{100}}_2$$

$$\underline{\underline{4^2 \rightarrow 16 \rightarrow}}$$

$$0 \dots \underline{\underline{10000}}_4$$

single set
bit

0/2/4/6 ... even spots.

num - 1 set bit

→ 0/2/4/6/8...

even pos.

0 → 31
 1 → 30
 2 → 29
 3 → 28
 ⋮
 31 → 0

$i \rightarrow (32 - i - 1)$
ans

$\xleftarrow{\quad} \xrightarrow{3} \xrightarrow{2} \xrightarrow{1} \xrightarrow{0}$
 n) 0 ... 0 1 0 1 1
 (... 0)

i bit 0/1
 content

n
 $\downarrow$
 $(32 - i - 1)$
 $\downarrow$
 ans

$\downarrow$
 ans = 1 1 0 1 (0 0 ... 0)
 = (0 1 1 0 ... 0)

$\textcircled{n} \rightarrow \frac{i}{0} (32 - i - 1)$

$i \rightarrow n \rightarrow \textcircled{1}$
 $\hookrightarrow (32 - i - 1) \hookrightarrow \textcircled{j}$

$$(32 - i - 1) \rightarrow \underline{j} \rightarrow (1) \xleftarrow{\text{ans}}$$

$$\underline{\text{ans}} \rightarrow \text{ans} \uparrow (1 < \underline{j}) \xrightarrow{\text{ans}}$$

$$i \rightarrow n \rightarrow (1) \uparrow (32 - i - 1)$$

0 → not present
1 → present

$n=3$ [1 2 3] → all subsets of 1/2/3

[] → 000
[1] → 001

[] [1 2]

[1] [1 3]

[2] [2 3] → bits

[3] [1 2 3]

[2] → 010

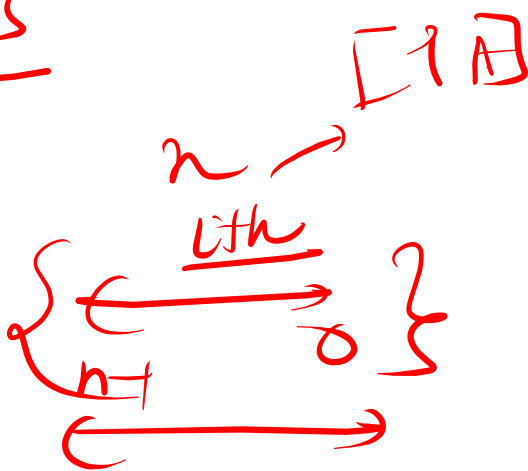
[3] → 100

[1 2] → 011

[1 3] → 101

[2 3] → 110

[1 2 3] → 111 → bit mask
⑦ 0-2ⁿ-1



2ⁿ

$$\underline{2^n} \rightarrow 0$$

$n \rightarrow [\underline{\text{subsets}}]$

$0 \rightarrow 000$
 $\vdots$
 $2^n - 1 \rightarrow 111$

} subsets
 of
 current
 array

$n=3$
 $\downarrow$

0	→	000	→ [123]
1	→	001	→ [1]
2	→	010	
3	→	011	
4	→	100	
5	→	101	
6	→	110	
7	→	111	

②

$n=2 \rightarrow \frac{10}{00} \rightarrow [2] \downarrow$

$i=0 \rightarrow \frac{01}{10} \rightarrow [2] \leftarrow \textcircled{1_2}$

$2^{n-1} \rightarrow 11 \rightarrow [12]$

$[] , [1] , [2] , [12]$

return

$123 \rightarrow [3] \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \end{pmatrix} \xrightarrow{\text{bit mask}}$

$[] = \begin{cases} 0 & 0 \rightarrow 0 \\ 0 & 1 \rightarrow 1 \\ 1 & 0 \rightarrow 2 \\ 1 & 1 \rightarrow 3 \end{cases} \Rightarrow \begin{cases} 0 \\ \vdots \\ 2^n - 1 \end{cases}$

$e \rightarrow \text{twice}$
single.
2
no.

single number !!!

1 2 1 3 2 5 2
←

~~1~~ ~~2~~ ~~1~~ ~~1~~ 3 ~~2~~ 5

↓ ↓ ↓
3 5
→ 3
→ 5

$(\underline{\underline{-110}})^2 \approx 2\text{'s complement}$

$$\begin{array}{r} 011 \\ 101 \\ \hline \Rightarrow 1 \text{ (1) } 0 \end{array}$$

fast set bit

2's complement

$$\begin{array}{r} 110 \\ \sim \downarrow \\ 001 \\ + 001 \\ \hline 010 \end{array}$$

$\xrightarrow{\text{num}}$

$$110 \times (010)$$

$$(0 \text{ (1) } 0)$$

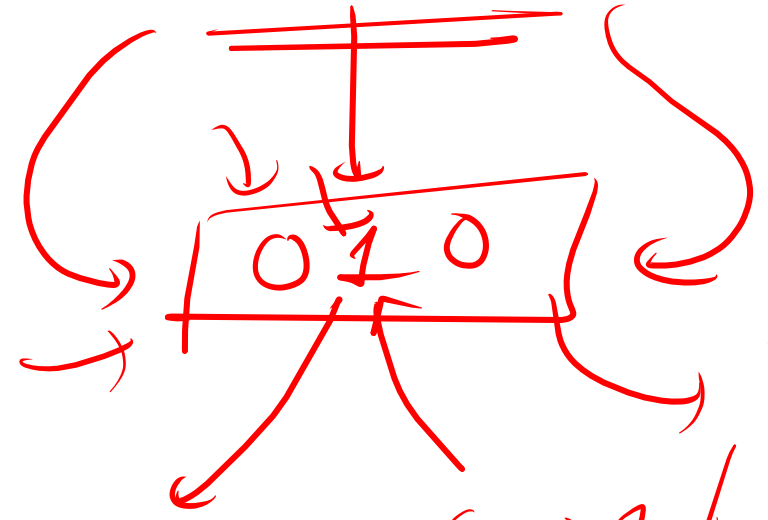
$$\boxed{110}$$

$$\begin{array}{r} 001 \\ + 001 \\ \hline 011 \end{array}$$

$$\begin{array}{r} 1111 \\ - 011 \\ \hline 1111 \end{array}$$

$\begin{array}{r} 3/2 \\ \hline 3/s_1 \\ 2 \quad 2 \\ 3 \end{array} \quad \bigg| \quad \begin{array}{r} 3 \\ \hline s_2/5 \\ 1, 1 \\ 5 \end{array}$

$\begin{array}{r} 2/13 \\ \hline 2/13 \\ 9_1 \end{array} \quad \begin{array}{r} 2/15 \\ \hline 2/15 \\ 9_2 \end{array}$
 $g_1 \rightarrow \underline{\underline{3}}$ $g_2 \rightarrow 5$

last set bit $\textcircled{X} \rightarrow s_1 \wedge s_2$
 $\rightarrow x \vee (-x)$ 2's complement

 $\text{signif} \rightarrow s_1 \rightarrow 1 / s_2 \rightarrow 0$
 $s_1 \rightarrow 0 / s_2 \rightarrow 1$

1 5 1 3 2 3 6 5

↓

① →

~~1~~ ~~5~~ ~~1~~ ~~3~~ ² ~~3~~ ⁶ ~~5~~

↓

2⁶ → 1 1 1 0

↪ 1 0 0

$$2's \rightarrow \underline{1111}0$$

$$2) \rightarrow \overset{2}{\circlearrowleft} \underline{\underline{100}} \vee (-100)$$

$$100 \vee (100)$$

$$\downarrow$$

$$\overset{2}{\circlearrowleft} \underline{\underline{100}} =$$

$$\overset{2}{\circlearrowleft} \underline{\underline{n}} \vee (-n) \rightarrow (\text{last set bit})^{0/>0}$$

$$\downarrow$$

$$\underline{\underline{10001000}}$$

$$\begin{array}{r} \sim 100 \\ 11 \\ 011 \\ + 001 \\ \hline 100 \\ \hline \end{array}$$

2's
complement

last set bit $\rightarrow \underline{\underline{100}}$

$$\begin{array}{r} 100 \\ 2011 \\ \hline 000 \end{array}$$

$$\begin{array}{r} 100 \\ \times 10 \\ \hline 000 \end{array}$$

15 13 2 36 5

$\rightarrow$ 21 $\rightarrow$ 1
 1, ~~5~~ ~~15~~ ~~26~~
 $\downarrow$
 $g_i = 6$

$f_2 \rightarrow 0$

~~$$\frac{1^2}{1^2 \cdot 2^2}$$~~

$\rho_2 = e^{-\rho_1}$

$$\begin{array}{r} 100 \\ 2 \overline{) 001} \\ \underline{000} \end{array}$$

$$\begin{array}{r} 100 \\ \times 101 \\ \hline (100) > 0 \end{array}$$

$$\begin{array}{r} 110 \\ \times 100 \\ \hline 1000 \end{array}$$

$x_{000}(\text{arr}) \rightarrow \frac{x_{000}}{2} \rightarrow 100$
 $\downarrow$
 s_1, s_2
 $\frac{s_1 \rightarrow 1}{s_2 \rightarrow 0} \} \text{vice versa}$

$$\begin{array}{cccccc} 2 & 2 & 2 & 2 & 2 & 2 \\ \hline 15 & 13 & 2 & 36 & 5 \end{array}$$

$$\begin{array}{r} 001 \\ \times 100 \\ \hline 000 \end{array}$$

$$\begin{array}{r} 011 \\ \times 100 \\ \hline 000 \\ 010 \\ \times 100 \\ \hline 000 \end{array}$$

$g_F = 1$
 $\frac{5}{16} \frac{15}{5}$
 $g_1 = 6$

$g_F = 0$
 $\frac{1}{13} \frac{1}{3} \frac{12}{2}$
 $g_2 = 2$

$$\begin{array}{r} 101 \\ \times 100 \\ \hline 1000 \end{array}$$

ans $\rightarrow [6, 2]$

three /
one → single

find

three times
 ↳ 2i + +3

single number |||

2 ✓ ✓ ✓
 { 1 2 3 1 2 1 2 }
 - 1 + 3 1 + 3 ✓
 1 0

3 → 0 1 1
 1 1
 + 1 + 1

single + 1 → bits

1 → ③

2 → 3

3 → 1

① → ② → ③
 ↙ 00 ① ↘

2 → 0 ① 0
 ↳ + 3

singly occ. ~~bit~~ number

$b_j \rightarrow +3+3+3$

$\underline{b_i} \rightarrow (1x) + \underline{(1)}$

$(3x)$

$(3x) / 3 = 0 //$

$b_i \rightarrow \underline{(3x+1)} / 3 = \underline{(1)}$

singly occ.
no.

$\underline{u} \dots \rightarrow 1 \rightarrow \text{remainder}$
 $\hookrightarrow \underline{\text{set bit}} \rightarrow \text{singly occ. element}$

$$\left\{ \begin{array}{cc} \frac{1}{4} & \frac{0}{4} \\ 1 & 1 \end{array} \right.$$

$0 \rightarrow 1$
 $1 \rightarrow 1$

singly
occ.
element

ans $\rightarrow$ $\left(\frac{1}{-} \quad \frac{1}{-} \right)$

ans $\rightarrow$ $\frac{0}{i}$

$$\left(\frac{1}{0} \quad \frac{1}{i} \right)$$

$$\left(\frac{1}{0} \right)$$

$\boxed{3}$
 4

10
10

2's complement

Java

-ve & 's
subtraction

-ve XORs

101
⊕ 010

111
↑
↑

$$5 - (2) =$$

$$\Rightarrow 101 + (10)$$

101 ?
101
01

10

4 bits

5 → 0101
2 → 0010

↓ 2's

~ → 1101

+ 1

1110

2's

5 → 32 bit → int
2 → 32 bit → int

101 + 1110

0011

positive

↓
0011 = 3

msb $\rightarrow$ ① $\rightarrow$ +ve

$\rightarrow$ val?

1111

0000

1

+

0001

0001

-ve no.

5 - 6

5 $\rightarrow$ 0101

6 $\rightarrow$ 0110 $\rightarrow$ 1001 $\rightarrow$ +1

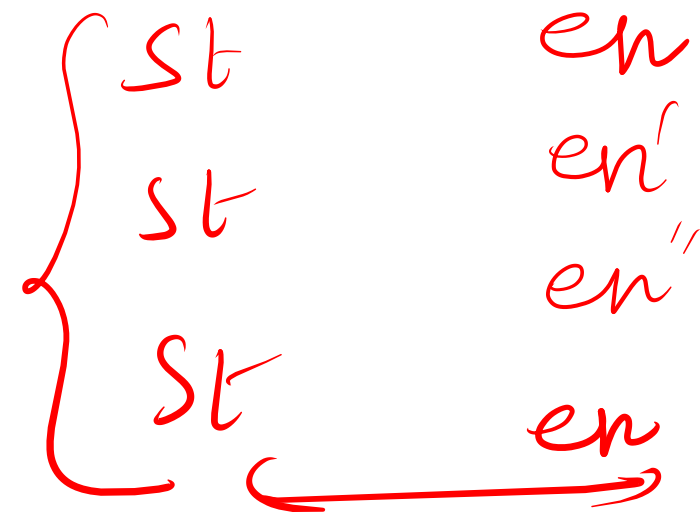
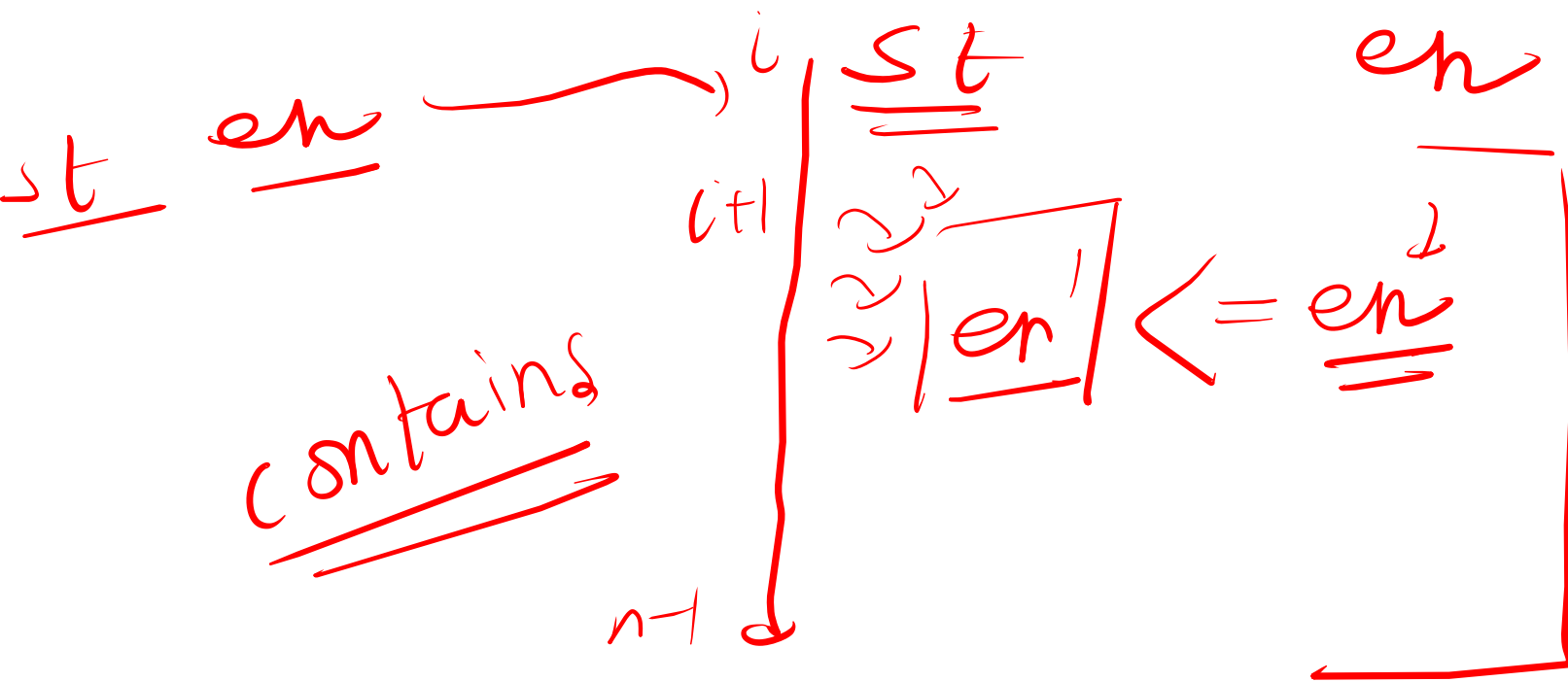
0101

+ 1010

1010

①111

15



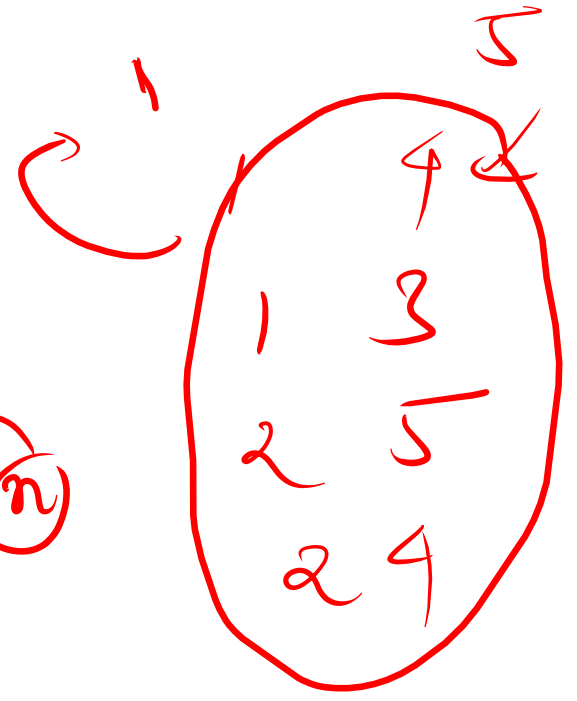
$\log(n)$ $\rightarrow$ seg/perm
PBDS

st en
 $\{ \text{en}' \leq \text{en} \}$
inanges
contains
ans

$\rightarrow 1 \rightarrow 5 \rightarrow 4$
 $\rightarrow 1 \rightarrow 4 \rightarrow 2$
 $\rightarrow 1 \rightarrow 3 \rightarrow 0$
 $\rightarrow 2 \rightarrow 5 \rightarrow 1$
 $\rightarrow 2 \rightarrow 4 \rightarrow 0$
 $\rightarrow 3 \rightarrow 7 \rightarrow 0$

$\rightarrow \{ 7, 4, 5, 3, 4 \}$
4

vector $\rightarrow$ $[\text{en} \text{ en}' \text{ en}''] \text{ } (n)$
 $\rightarrow (i)$
 $O(n * n)$



c st en
 $\begin{array}{c} 14 \\ \hline \end{array}$

$st \leftarrow 1$
 $\hline \hline$

st
 $\downarrow$

$en = 4$
 $\hline \hline$

$0 \leftarrow c$
 $1 \leftarrow 14$
 $\hline$

$13 \rightarrow 2$
 $\hline$

$25 \rightarrow 1$
 $\hline$

$\rightarrow 24 \rightarrow 3$

$\rightarrow 37 \rightarrow 0$
 $\hline$

$\{ \overset{2}{\overset{2}{5}} \overset{2}{4} 3 5 4 \}$
 $\hline$

$\downarrow$

vector

$O(n^2)$

$O(\log(n))$

GNU
 $\hline \hline$

$\rightarrow \underline{\underline{PDS}} / \underline{\underline{gen}} / \underline{\underline{seq}}$

```
#include <bits/stdc++.h>
```

```
#include <ext/pb_ds/assoc_container.hpp>
```

```
#include <ext/pb_ds/tree_policy.hpp>
```

```
using namespace __gnu_pbds;
```

```
template<typename T>
```

```
using ordered_multiset = tree<T, null_type, less_equal<T>, rb_tree_tag,  
tree_order_statistics_node_update>;
```

```
int main() {
```

```
    ordered_multiset<int> s;
```

```
    s.insert(5);
```

```
    s.insert(5);
```

```
    s.insert(7);
```

```
    s.insert(10);
```

```
    // find_by_order: 0-based
```

```
    std::cout << *s.find_by_order(0) << "\n"; // 5
```

```
    std::cout << *s.find_by_order(2) << "\n"; // 7
```

```
    // order_of_key: count of elements strictly less than value
```

```
    std::cout << s.order_of_key(6) << "\n"; // 2 (two 5s are < 6)
```

```
}
```

int/float

{ 4:3 }

{ 4 4 4
5 5 5 }



0 1 2
└─────────┘

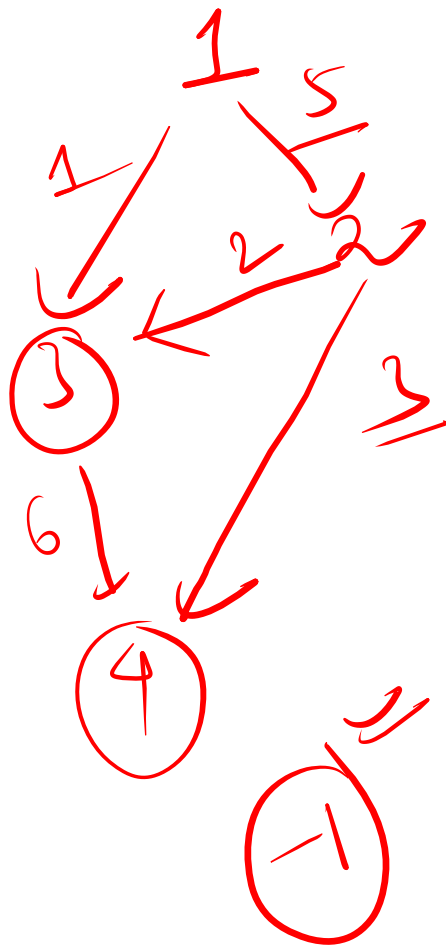
<6

path?

$$\frac{U}{V} = \frac{V}{V}$$

~~K edge path~~

max


$$t = 7$$

$K=2$

$\textcircled{X} 1 \cdot 3 \cdot 4 \rightarrow 7$
 $\textcircled{X} 1 \cdot 2 \cdot 3 \rightarrow 7$
 $\textcircled{X} 1 \cdot 2 \cdot 4 \rightarrow 8 \textcircled{X}$
 $\textcircled{X} 1 \cdot 2 \cdot 4 \rightarrow 8$
 $2 \cdot 3 \cdot 4 \rightarrow 8 \textcircled{X}$

src

0 --- n-1

node

k edges.

ans

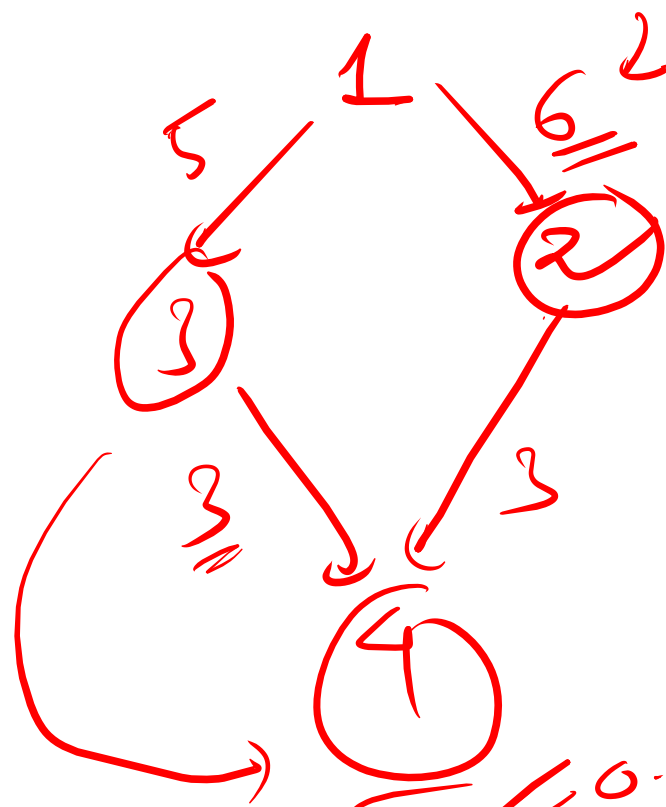
max.

node

parent

0 1 2 3 ... k

{dte}

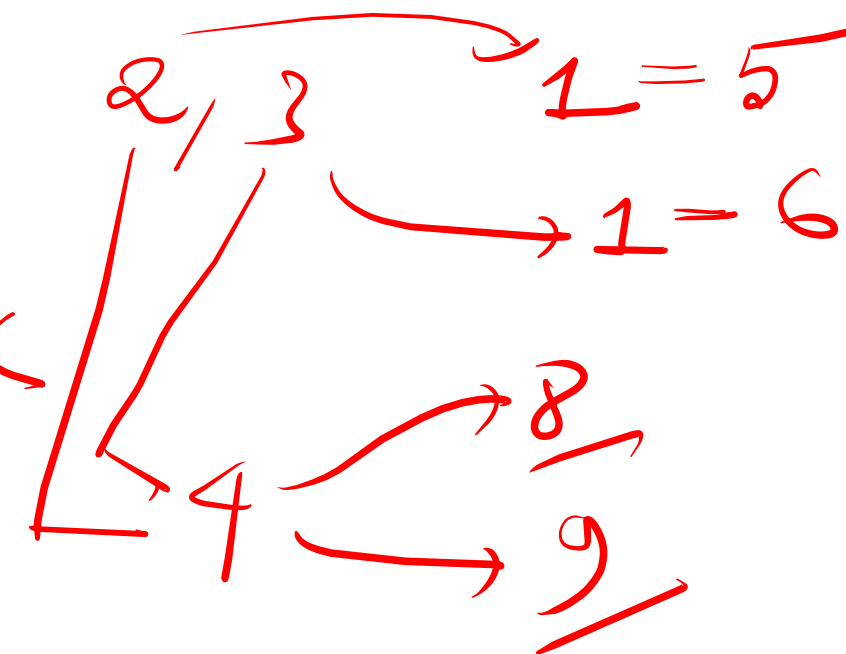


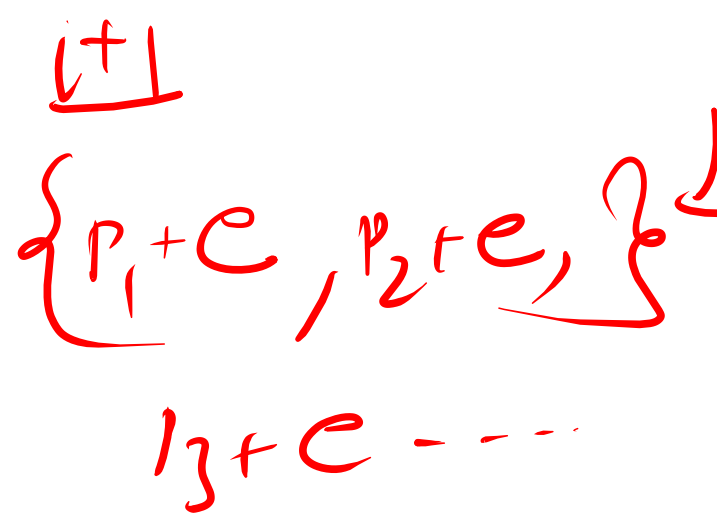
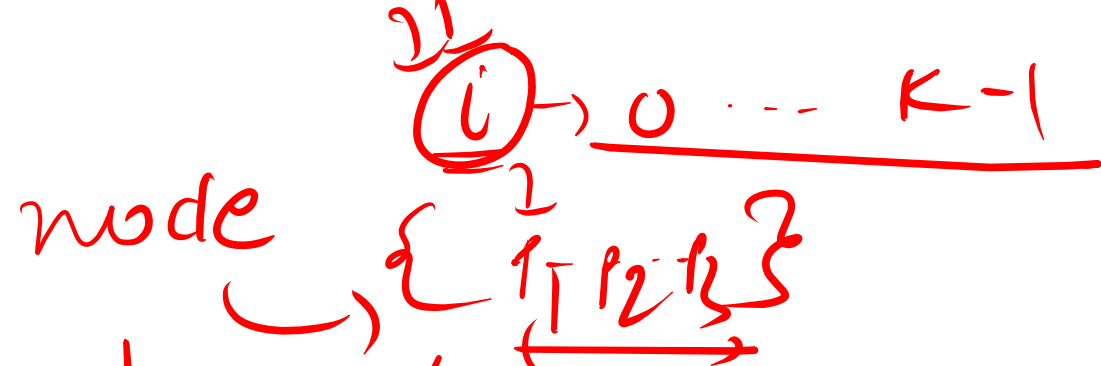
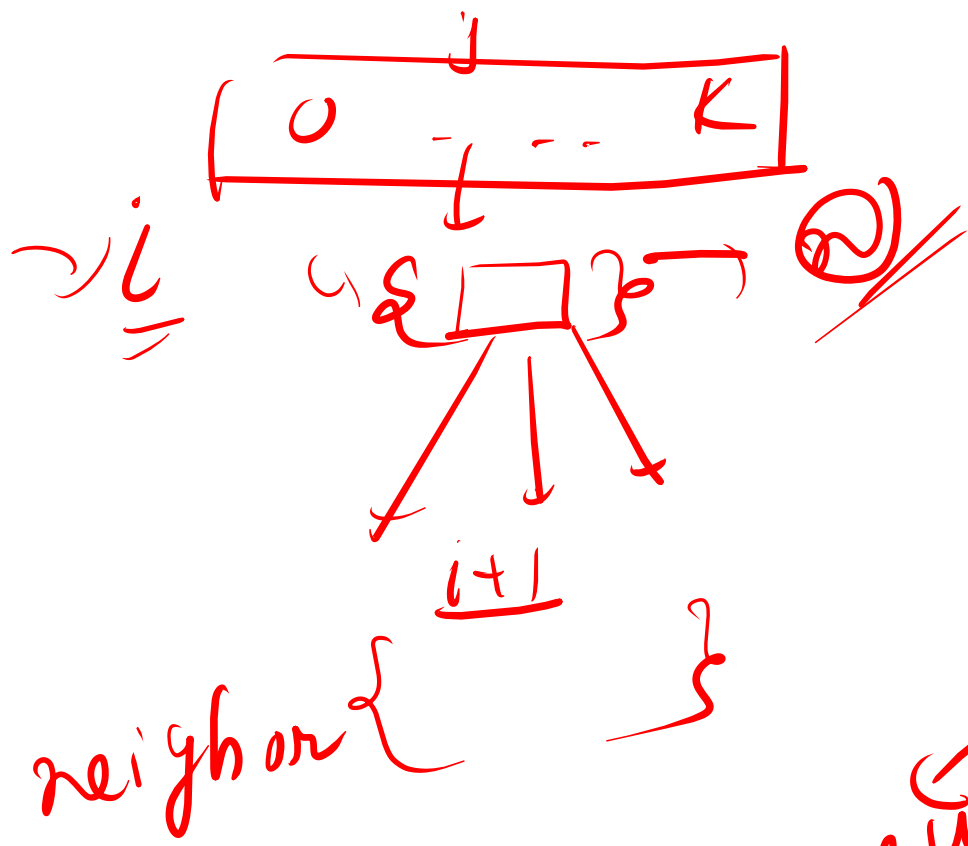
~~t=10~~ 9
K=2

1 → 0

1 = 5

1 = 6





dfs/bfs
distance
+ dp

union find / union rank
✓ ✓

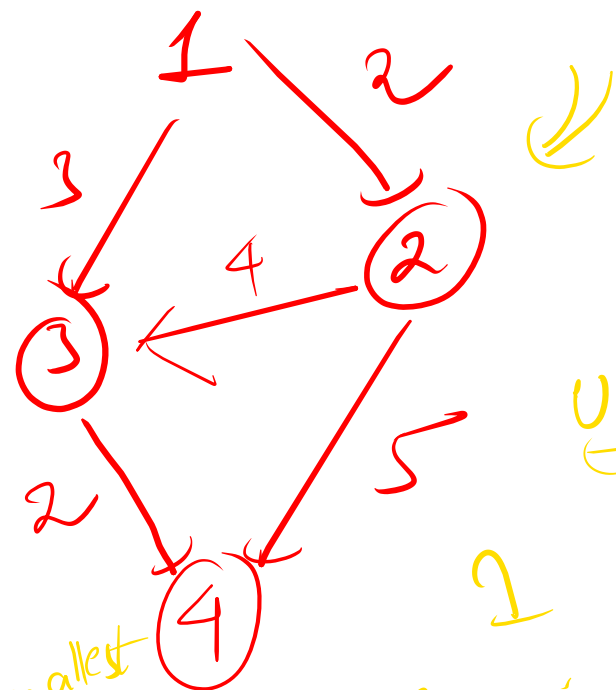
min possible distance

Floyd warshall algo

$\infty \rightarrow INT_MAX$

req.

all pair shortest path



	1	2	3	4
1	0	2	3	5
2	∞	0	4	5
3	∞	∞	0	2
4	∞	∞	∞	0

all grid $\rightarrow \infty$

4

s

$\rightarrow$ smallest

$K \rightarrow 1$

for($k=1; k \leq n; k++$) {

for($u=1; u \leq n; u++$) {

for($v=1; v \leq n; v++$) {

dist[u][v] =

min(dist[u][v],

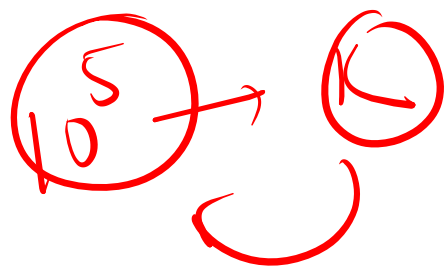
dist[u][k] + dist[k][v]);

} }

$O(n^3)$

$n \approx 100$
DP

≈ 1000



{ segment
tree }

pref i

$\rightarrow [1, 2, 4, 5, 6, 7] \rightarrow \{ \overset{2}{\overset{6}{10}} \}$

ans
return
pref [2]
pref [2-1]

$\rightarrow Q \rightarrow$

(L R) $\rightarrow$ subarray sum

idx
prefix sums

O(4)

Q

$$\frac{R-L}{2} \rightarrow \underline{\underline{n^2}}$$

$$Q \rightarrow \underline{\underline{O(N)}}$$

$$N \rightarrow 10^6 / Q \rightarrow 10^5$$

$$\frac{10^8}{10^{11}}$$

$$(10^5 * \underline{\underline{10^6}})$$

$$\left[1 \ 5 \ \overset{2}{\underline{\underline{2}}} \ \overset{4}{\underline{\underline{6}}} \ 4 \right] \rightarrow \begin{matrix} \text{gcd} \\ \underline{\underline{\text{ans}}} \end{matrix}$$

$$Q \rightarrow \begin{matrix} L & R \\ \hline \end{matrix} \downarrow \text{gcd?}$$

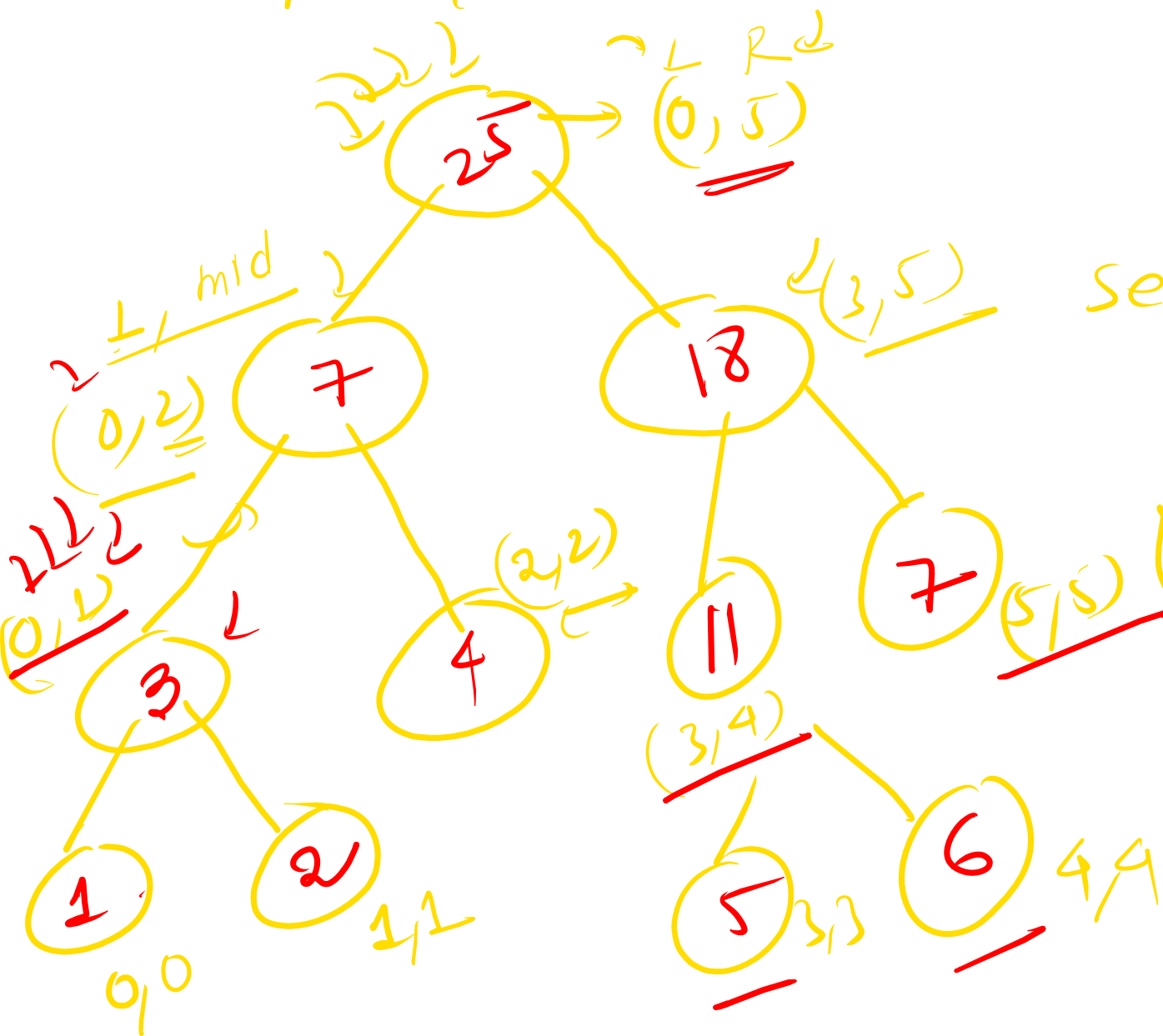
$$\begin{matrix} \text{---} \\ L & R \\ \text{---} \end{matrix} \downarrow \text{gcd} \rightarrow \underline{\underline{\text{ans}}}$$

$$\text{Query} \rightarrow \log(N) \downarrow \text{segment trees}$$

$n=6$
 ✓ 1 2 4 5 6 7 sum

tree?

Nodes & edges



segment

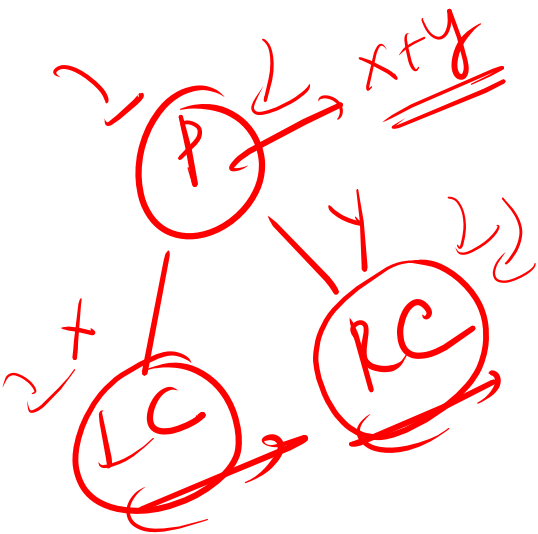
sum
ged

nodes.

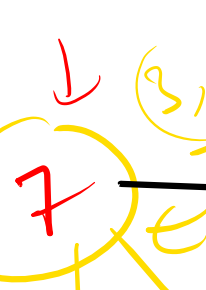
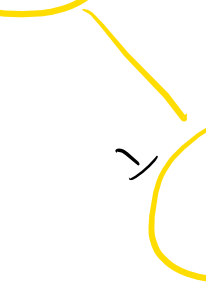
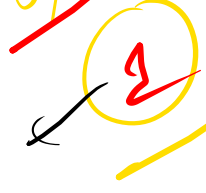
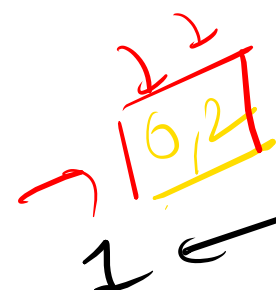
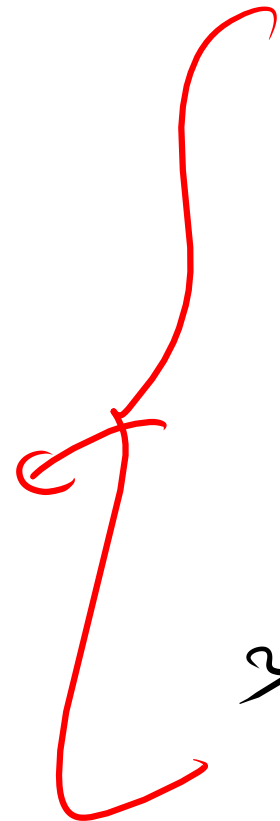
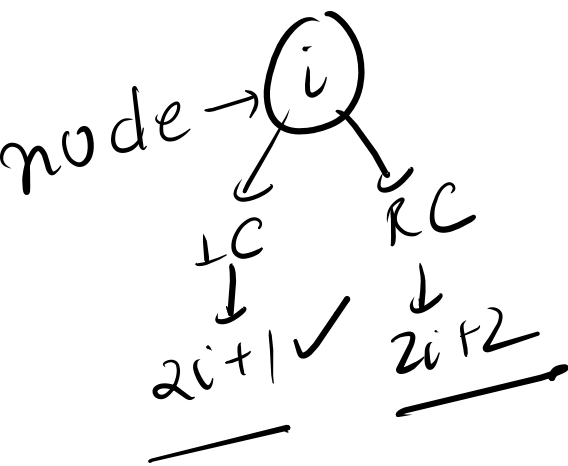
seg
faultt

1 5 2 3 4 $\rightarrow$ (0--4)

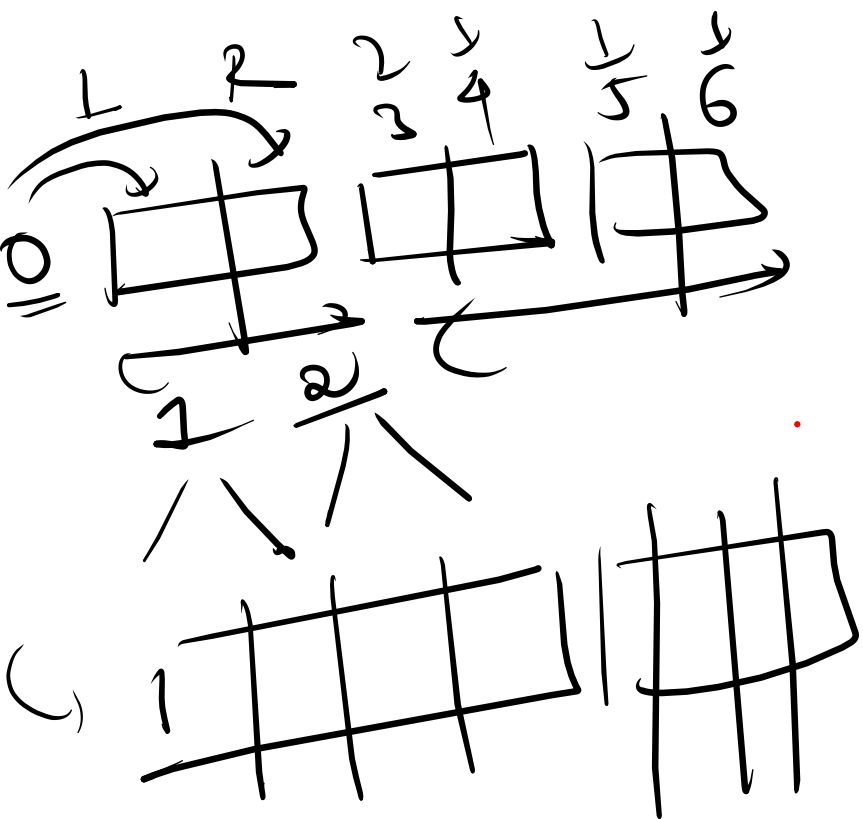
$\textcircled{n}$
 $\hookrightarrow 4n \rightarrow$ safe
 limit



arr

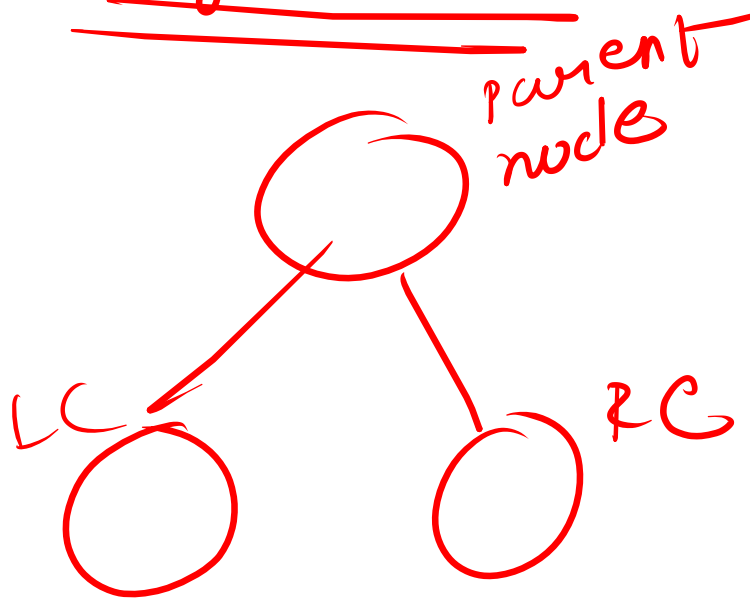


st, en
 $(st == en)$
 $\downarrow$
arr[st]

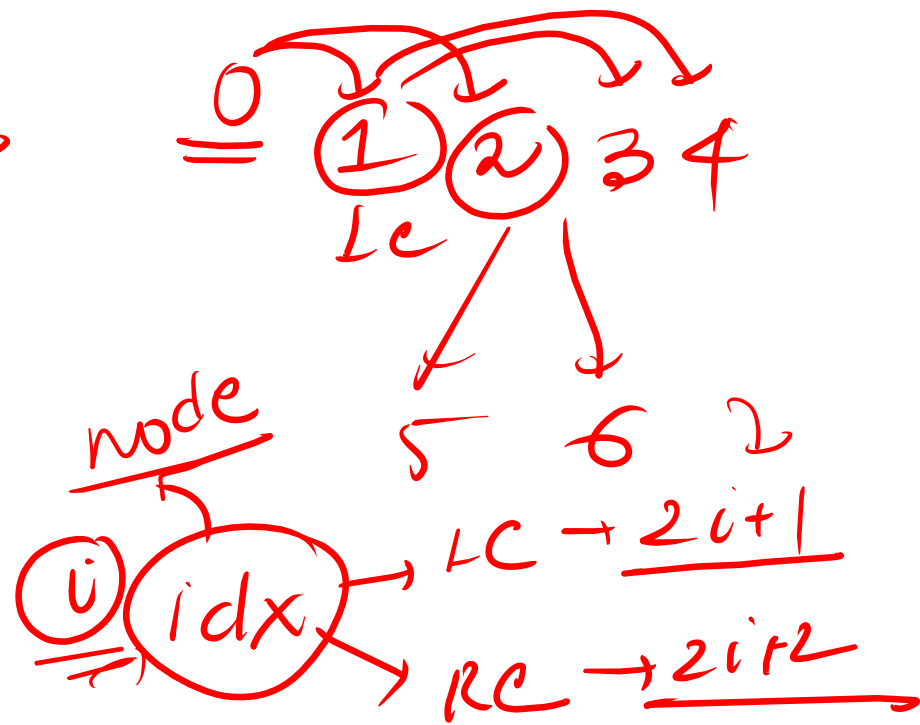


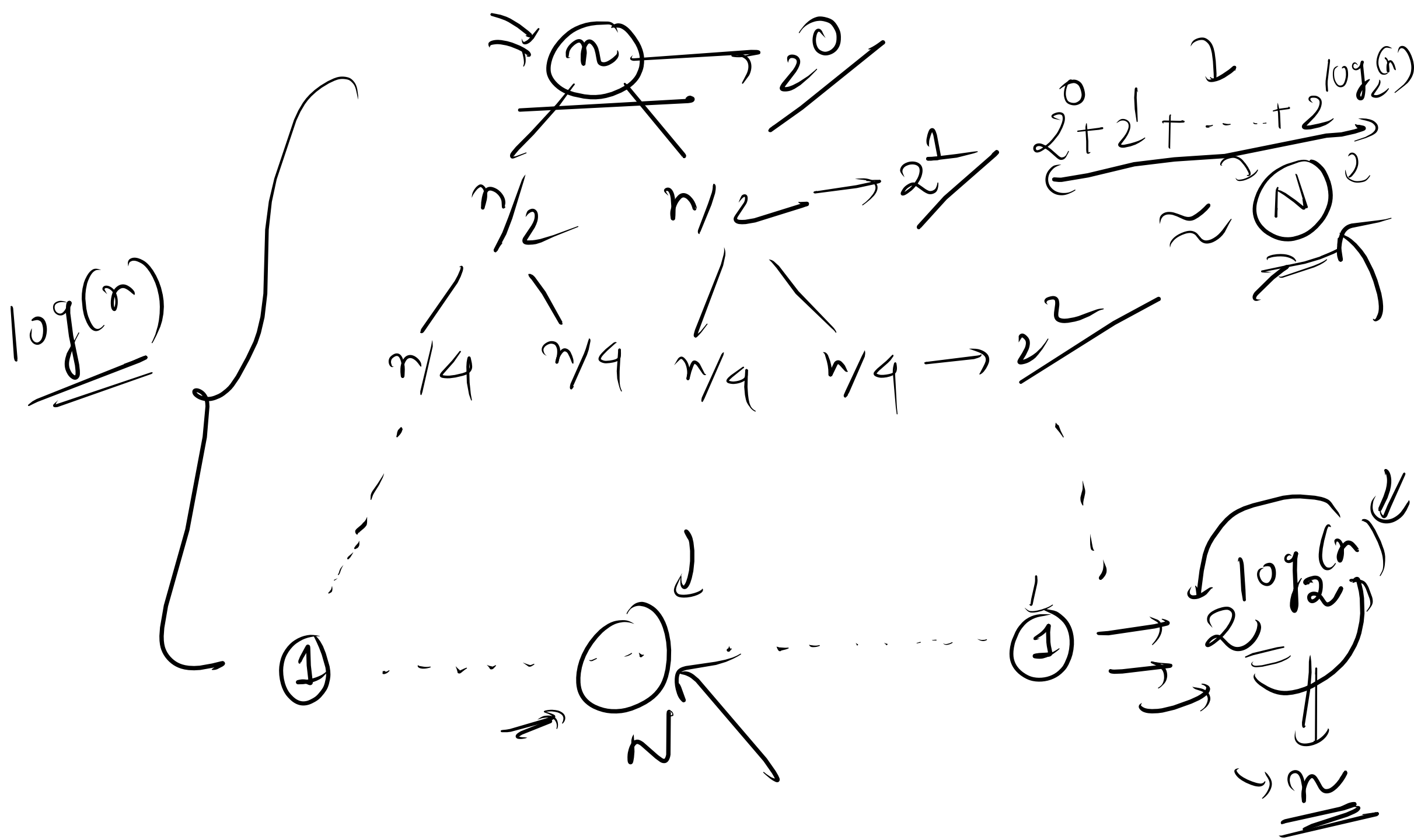
$$i \rightarrow \begin{matrix} \text{LC} \rightarrow 2i+1 \\ \text{RC} \rightarrow 2i+2 \end{matrix}$$

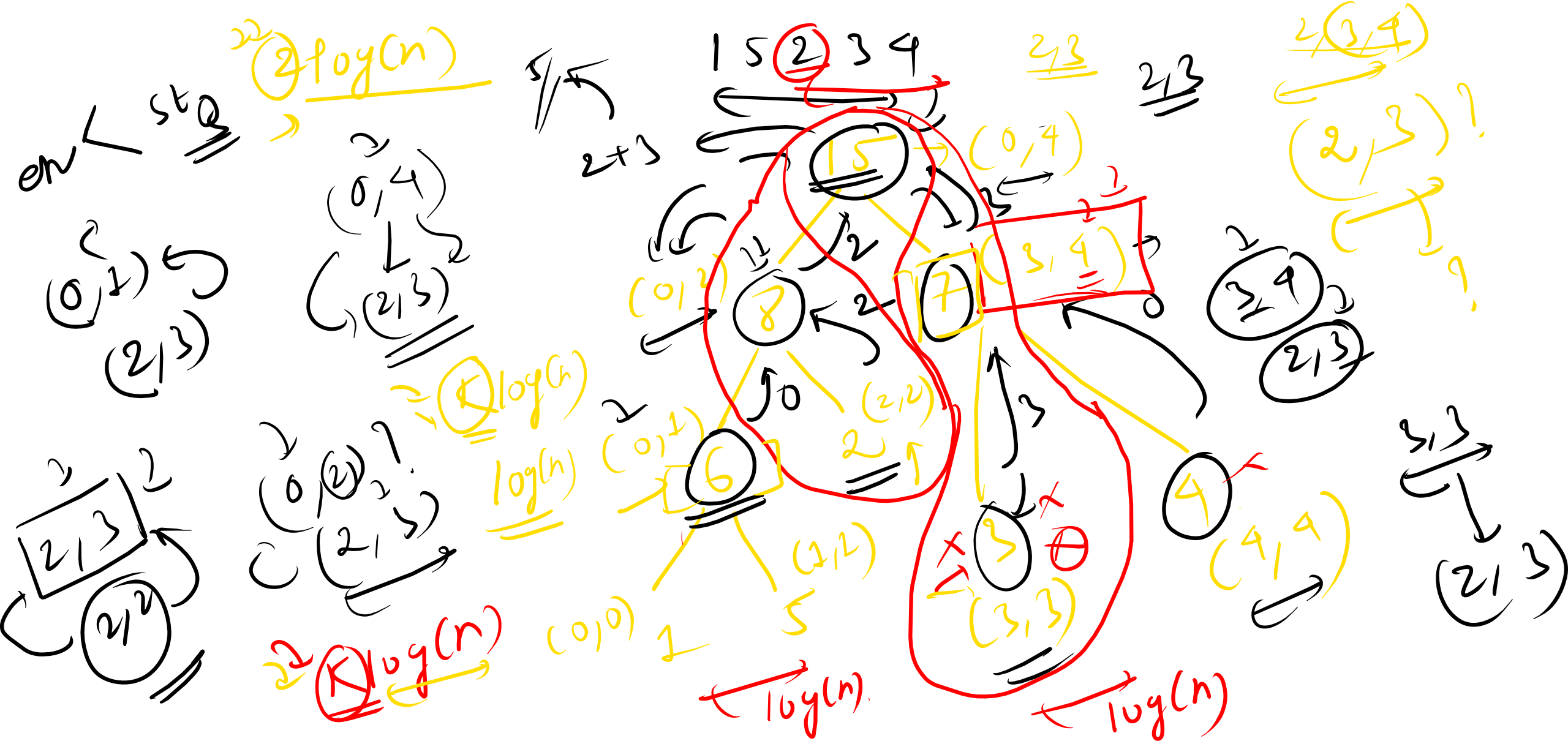
seg tree? $\rightarrow$ heaps? $\leftarrow$



arrays?







$$2n^5$$

$$(n^2)$$

$$h \rightarrow 10^5$$

0

→

→

→

→

$\log(n)$

→ seg faults

$$\{ 4n \}$$

$$\{ 1-9 \}$$

idx

← complete overlap →

seg[idx] →

complete non-overlap
↓

return 0;

partial overlap

(L) (R)
↘ ↙

→ return

seg?
 $O(\log(n))$
[Diagram showing overlapping segments with arrows indicating a search process]

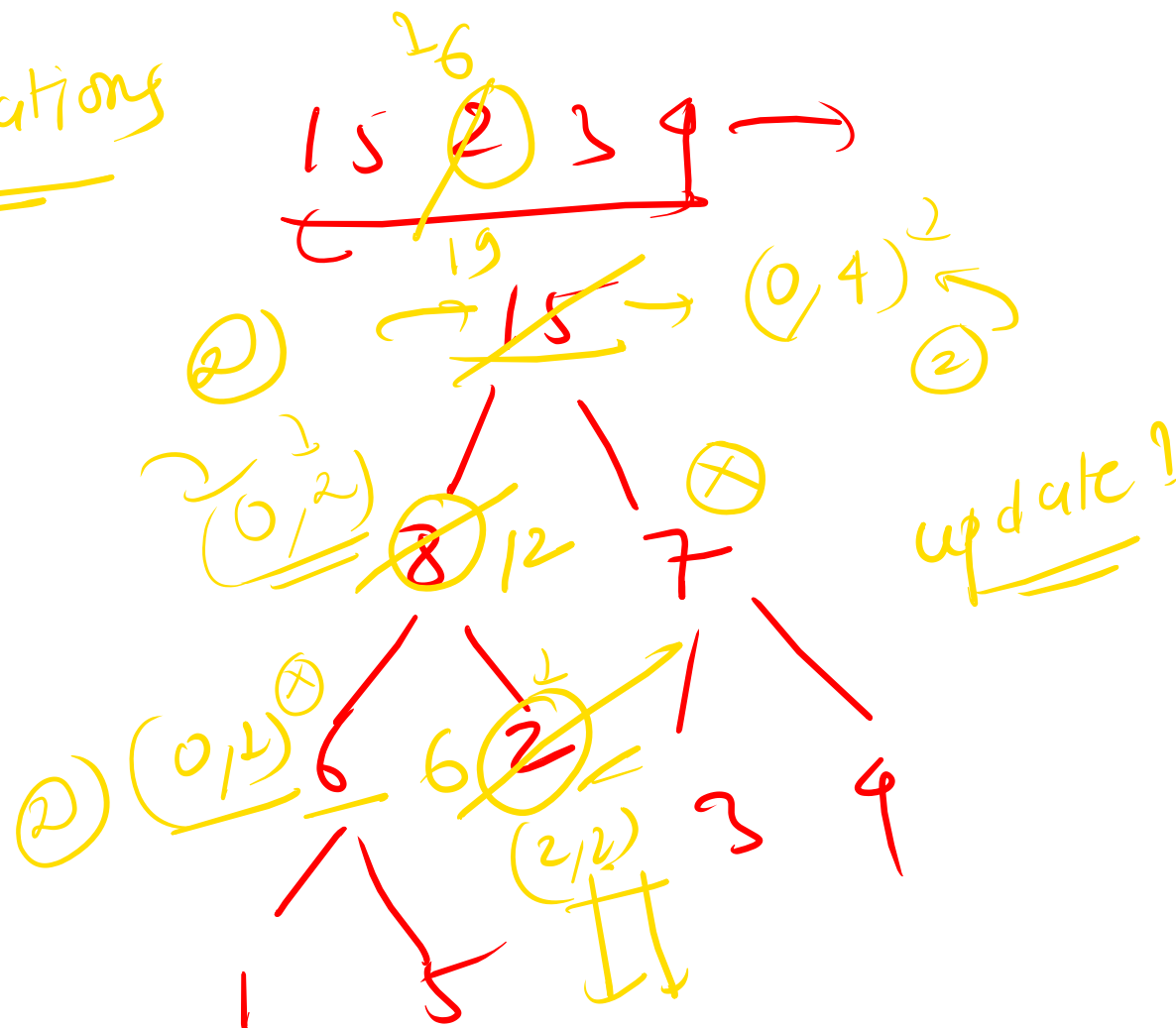
applications

update

root

$\log(n)$

$\log(n)$

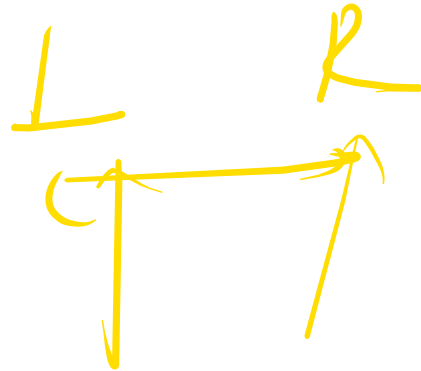


lazy propagation

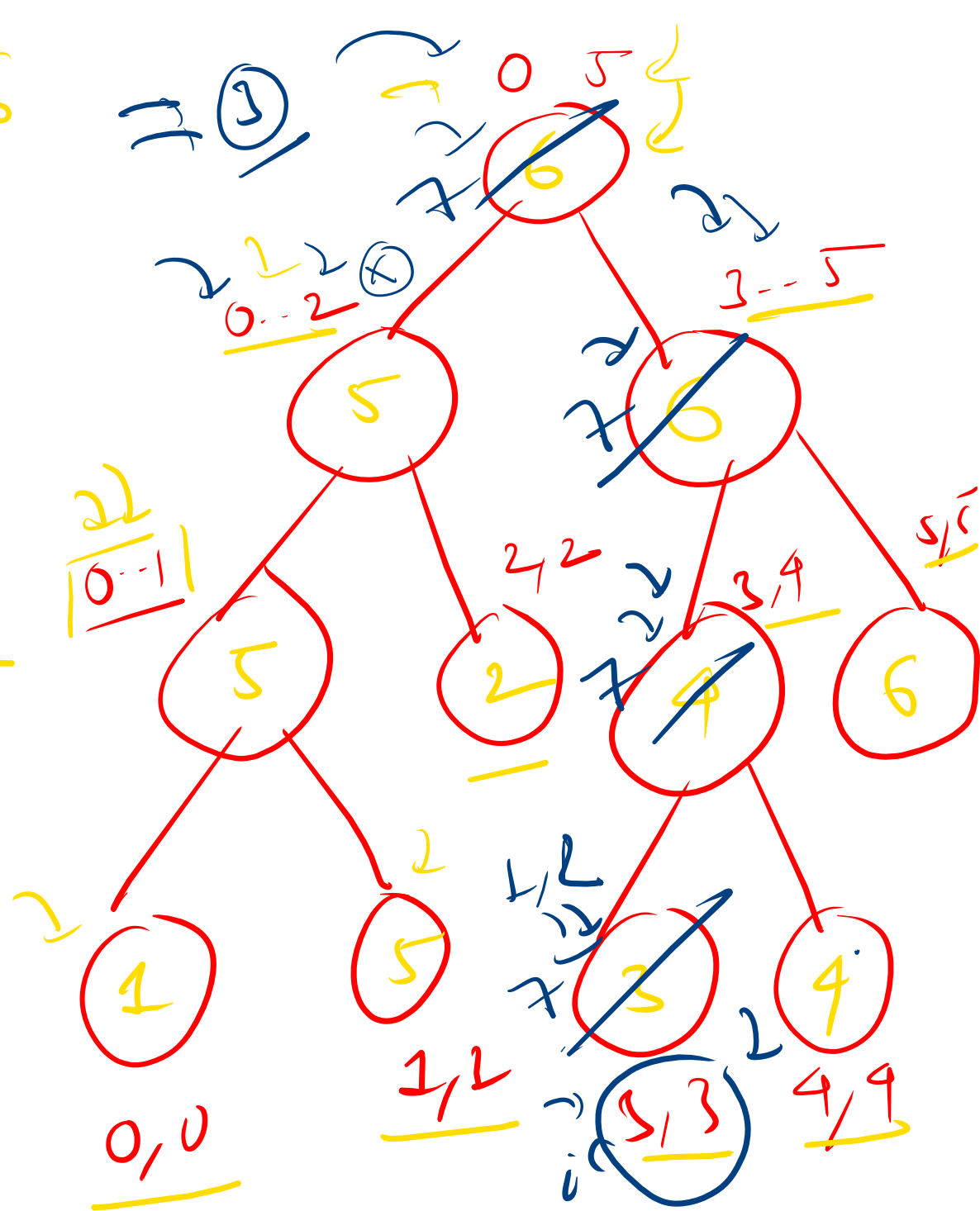
arr $\rightarrow$ 1 5 2 ~~3~~ 4 6

seg tree

Q



indices



1 5 5 1 1 2 3 6 9 $\rightarrow$ (pair)
 $\downarrow$
 $\rightarrow (L, R) \rightarrow (\underline{\text{max}} / \underline{\text{freq}}) = ?$
 $\downarrow$
max $\rightarrow$ freq

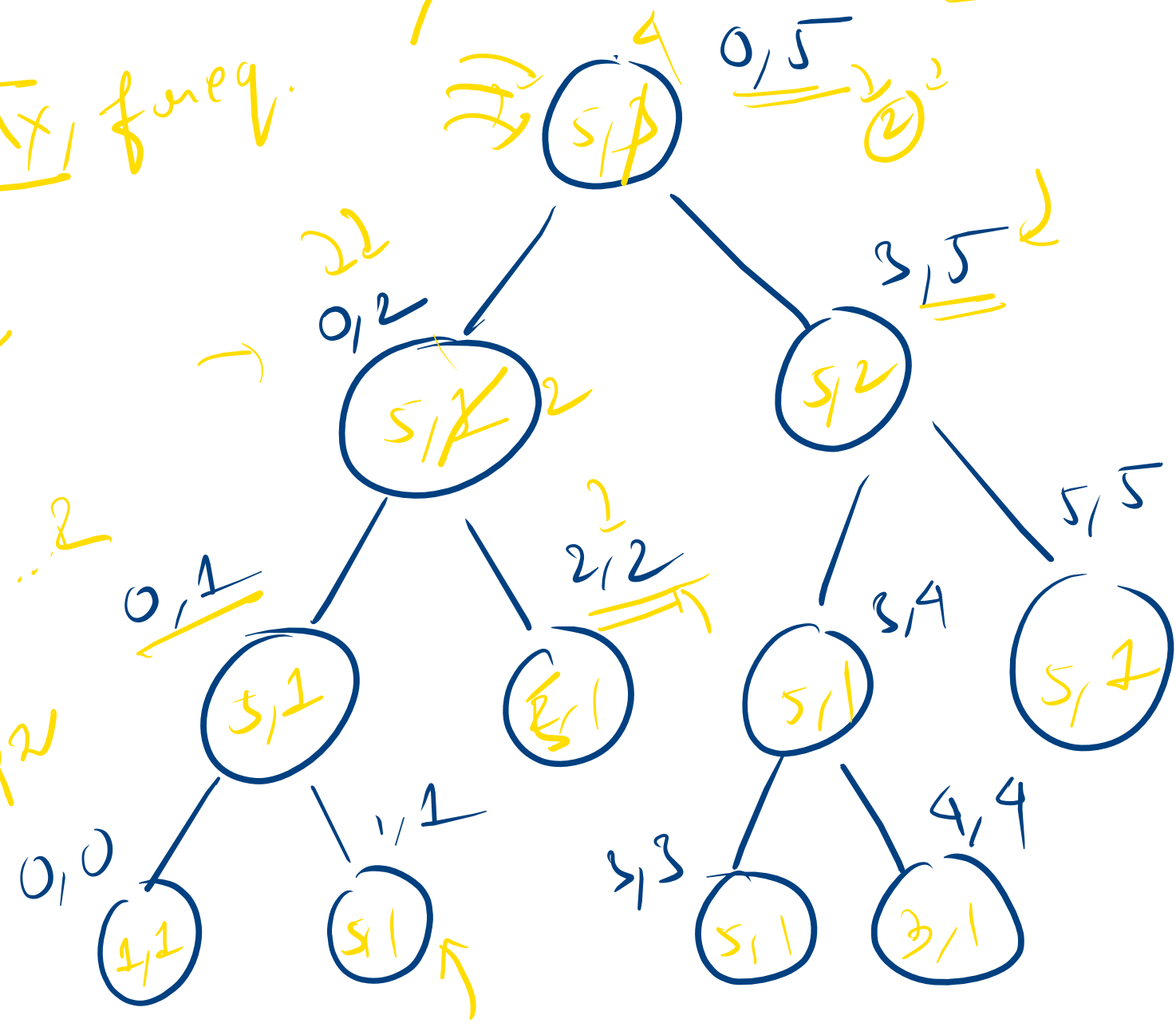
1 5 2 5 3 5

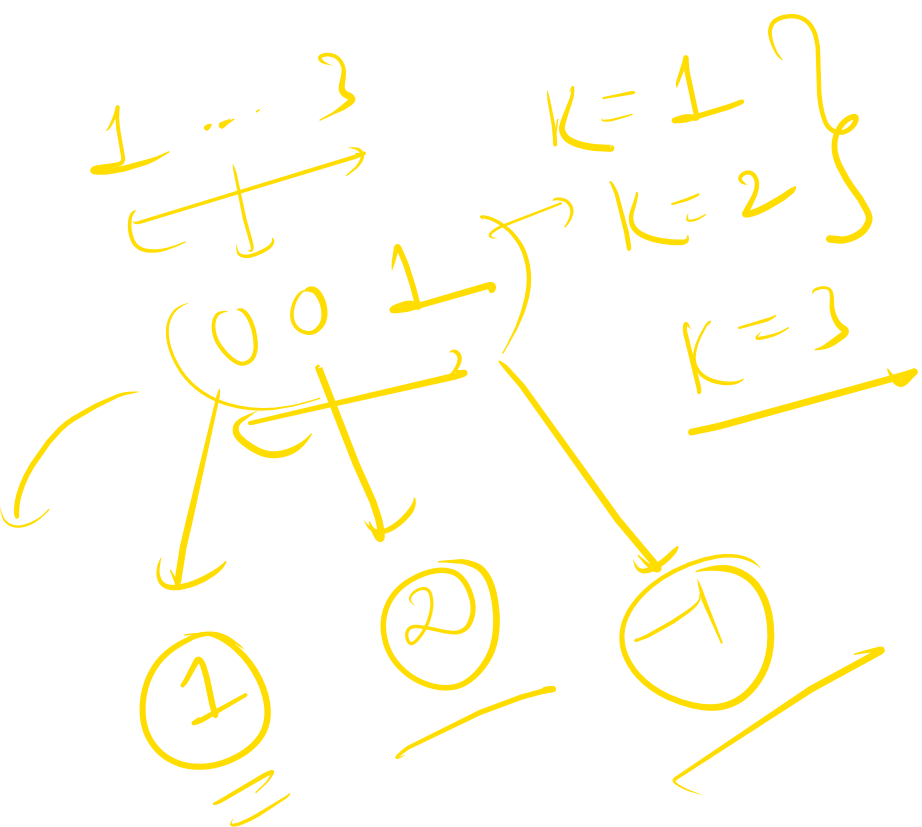
L...R $\rightarrow$ map $\checkmark$

max freq.

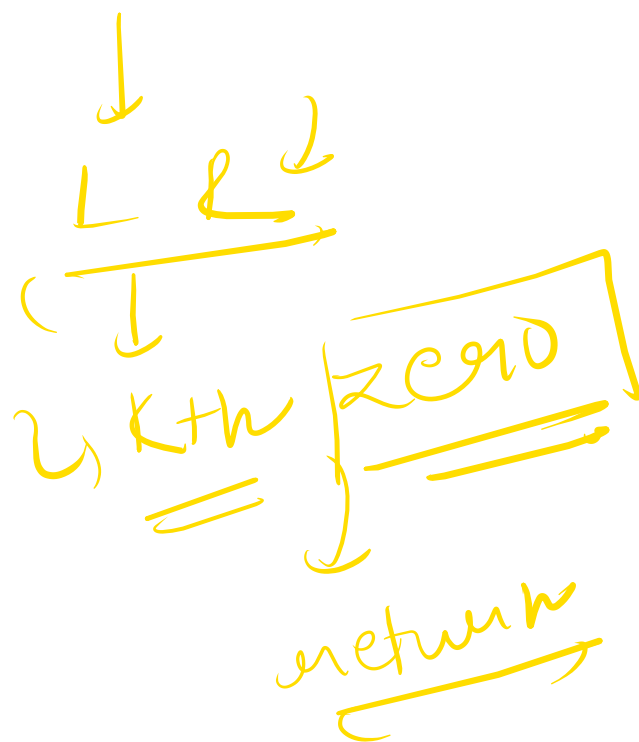
(22) $\rightarrow$

map1 $\rightarrow$ 1...mid
mid+1...2
map2
map $\checkmark$



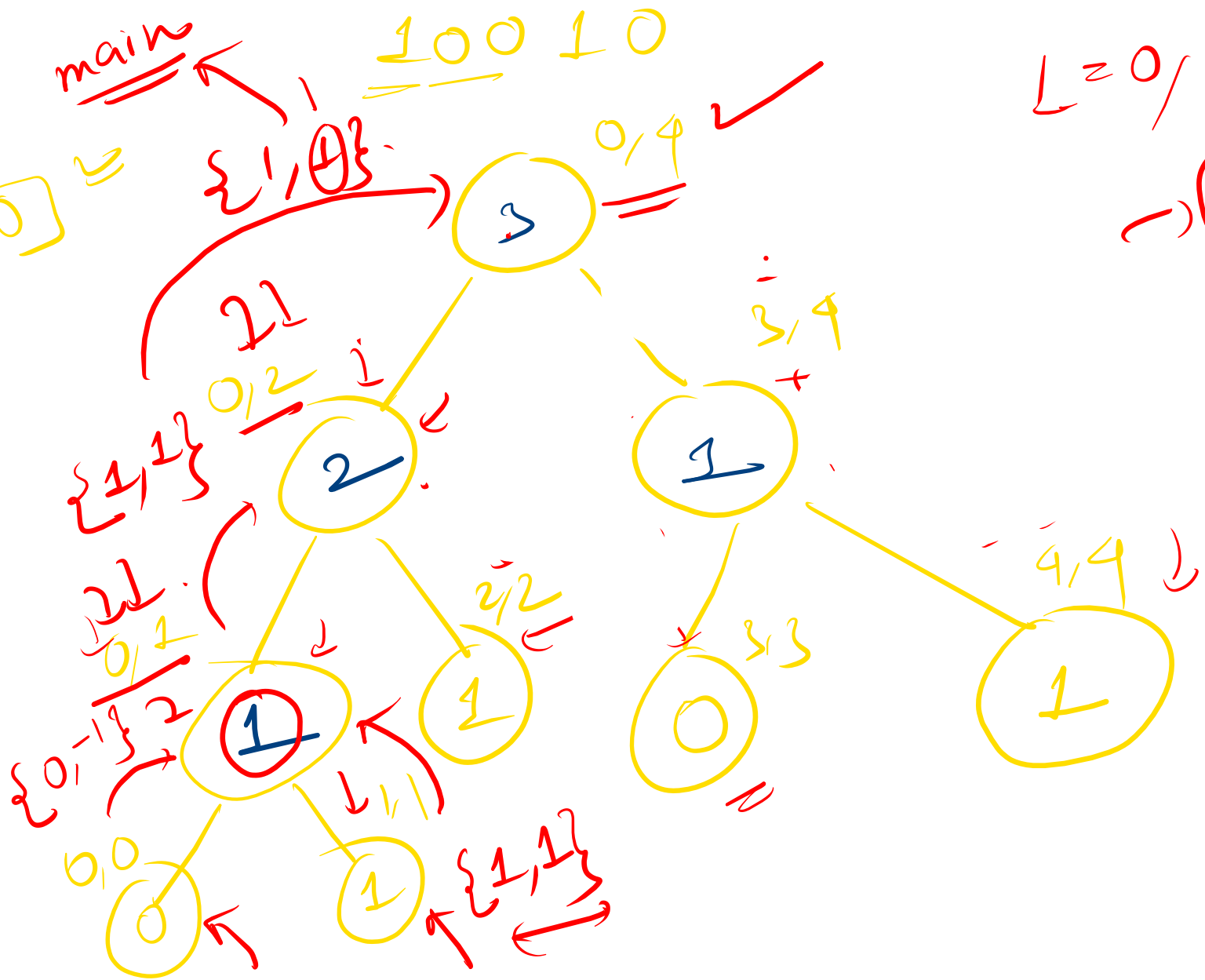


1 0 0 1 0



$H \rightarrow L \dots R$
 $\downarrow$
 count $\rightarrow 10$

$\{ \text{count, last} \}$
 $\rightarrow ?$
 $0, -1$



$L=0, R=2$
 $\rightarrow (K=1)$
 $\geq K$

1, 4, $K=3$

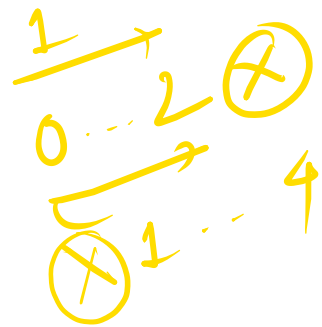
$\{2, 2\}$



10010

$0, 2, K=1$

{count, last}



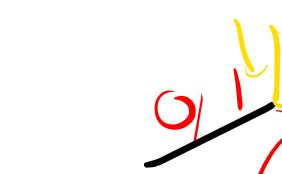
$3 > K$

$2 > K$

$2 > K$

$1 = K$

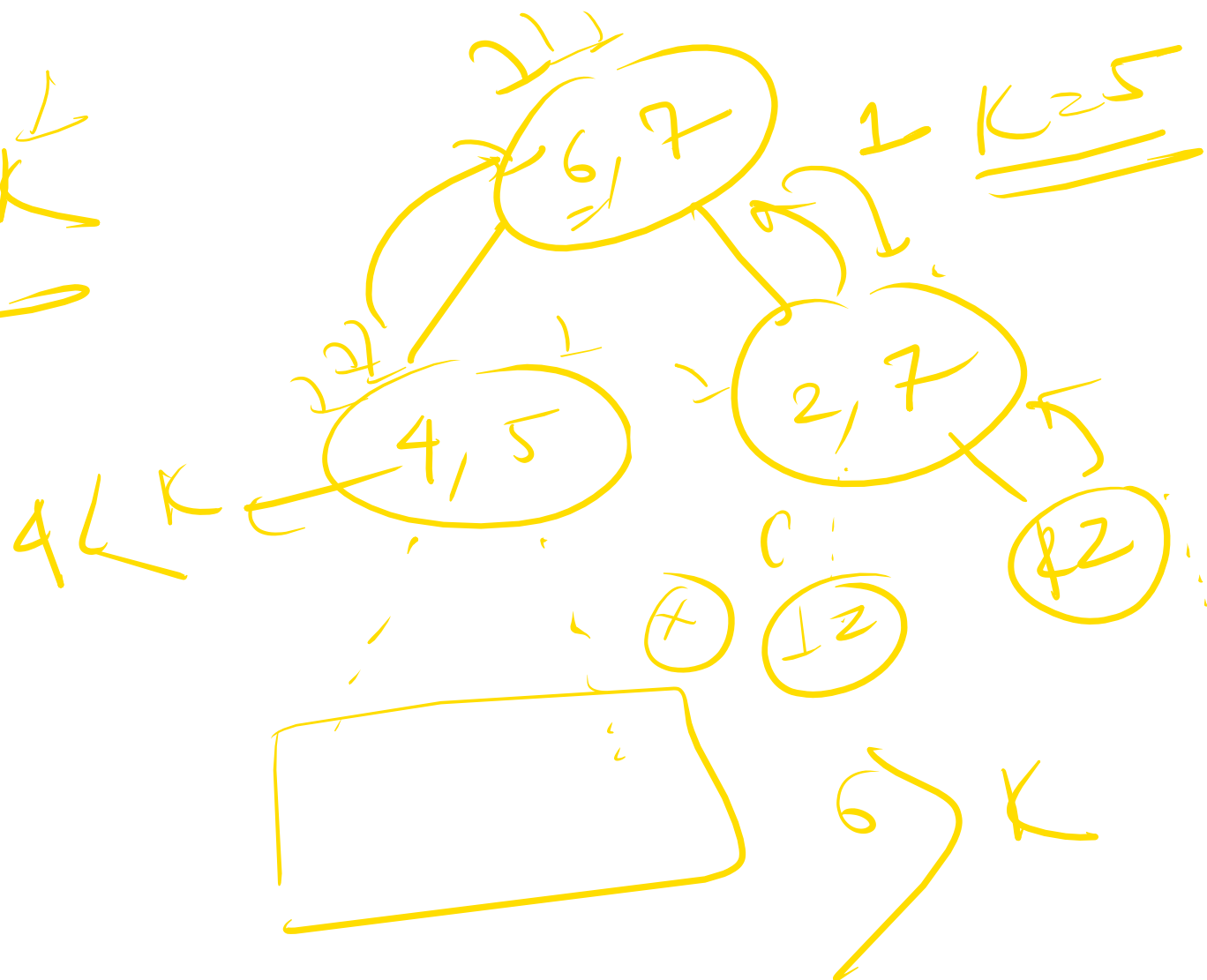
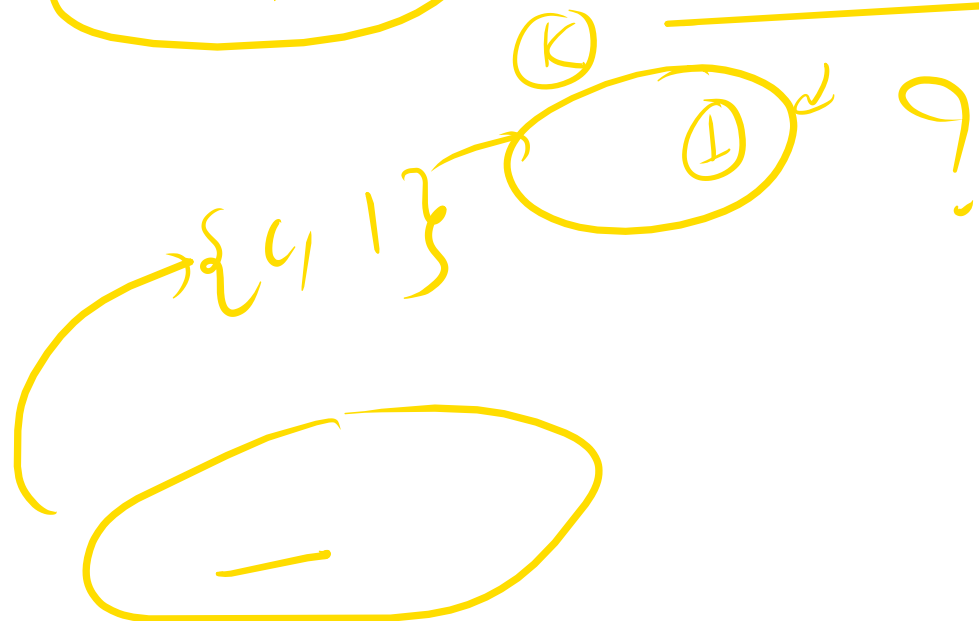
27^K



$\{2, 2\}$

$1 = K$

$\overbrace{L, R} \rightarrow$
 count, last
 st/en
 $\text{count} < K$



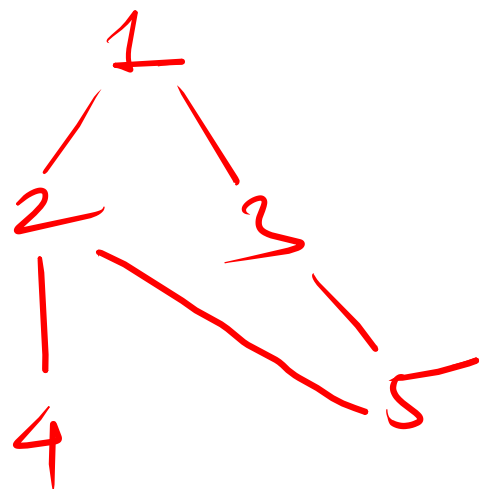


merge!

p_1 p_2

$\checkmark$

$\#$

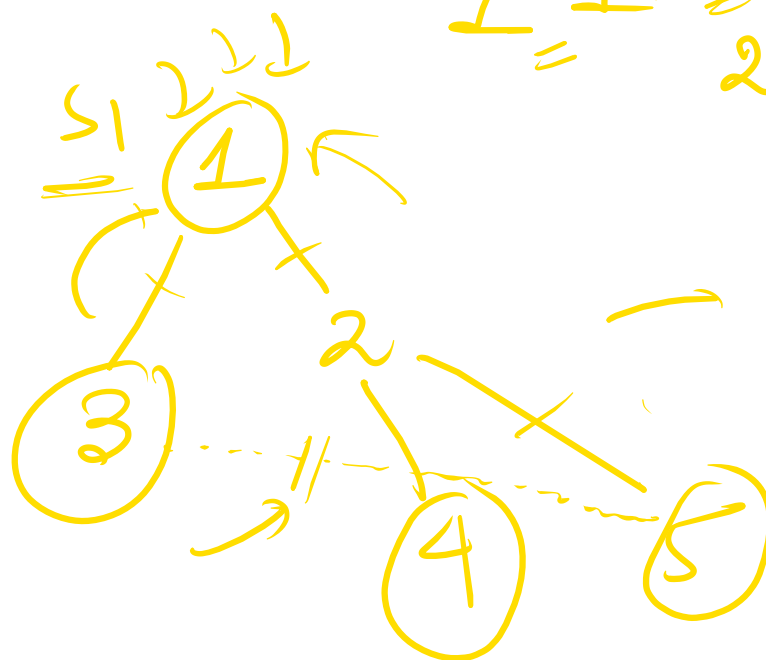


edges.

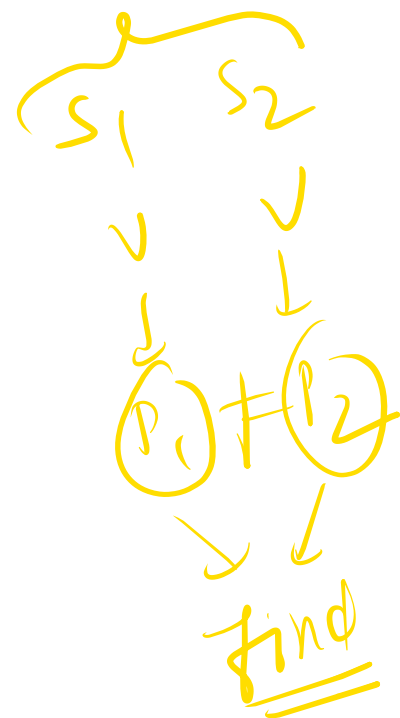
$\left\{ \begin{array}{l} \underline{1-2} \\ \underline{2-4} \\ \underline{1-3} \\ \underline{2-5} \\ \underline{3-5} \end{array} \right\}$

Union

$\checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark$
 $P: \quad \underline{1} \quad \underline{2} \quad \underline{3} \quad \underline{4} \quad \underline{5}$
 $\quad \quad \underline{1} \quad \underline{1} \quad \underline{2} \quad \underline{2}$

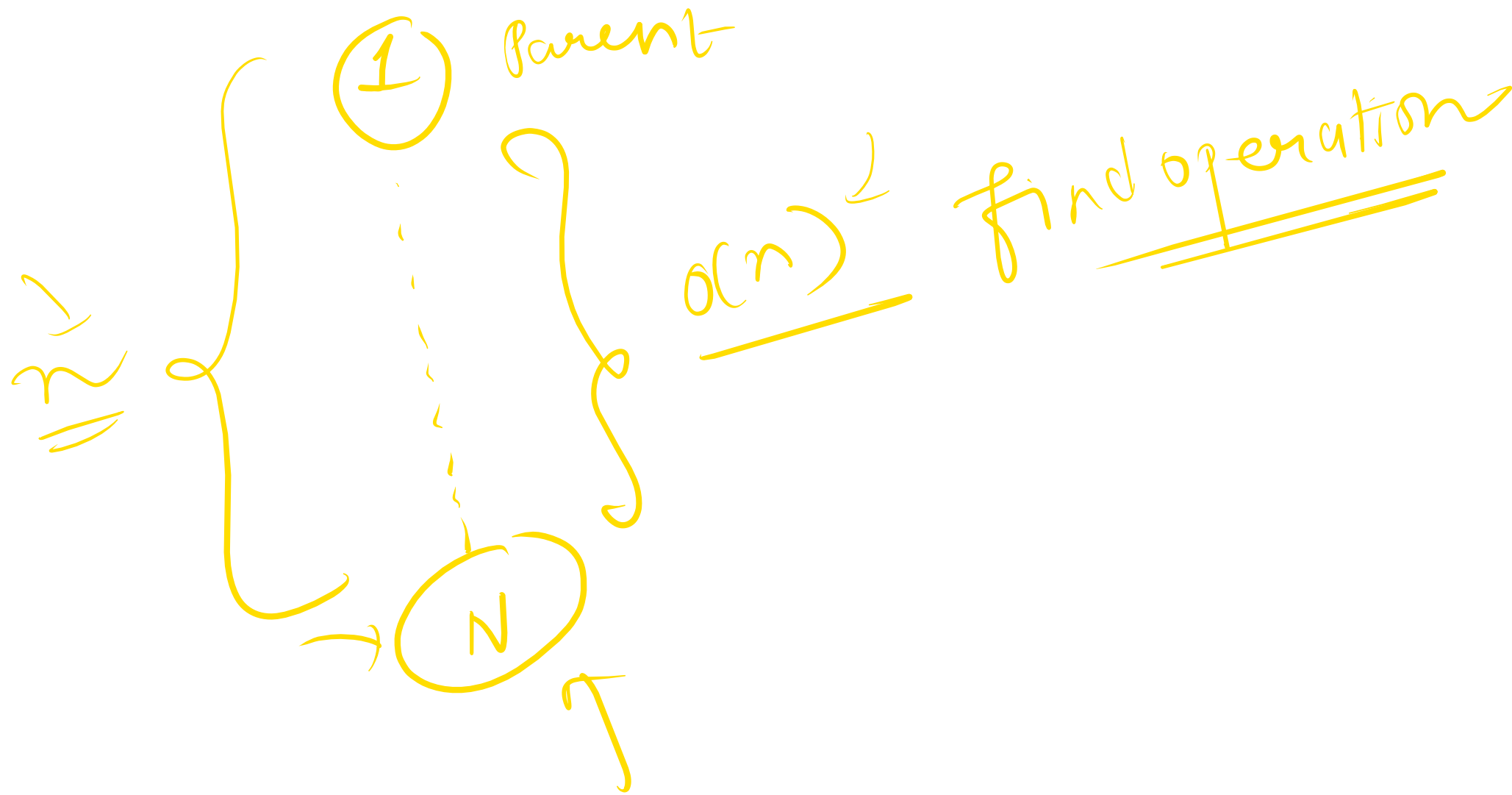


union.



??

cycle detection



$1-2 \checkmark$
 $2-4 \checkmark$
 $3-5 \checkmark$

$6-7$
 $8-9$

$2-6$

$3-7$

$7-9$

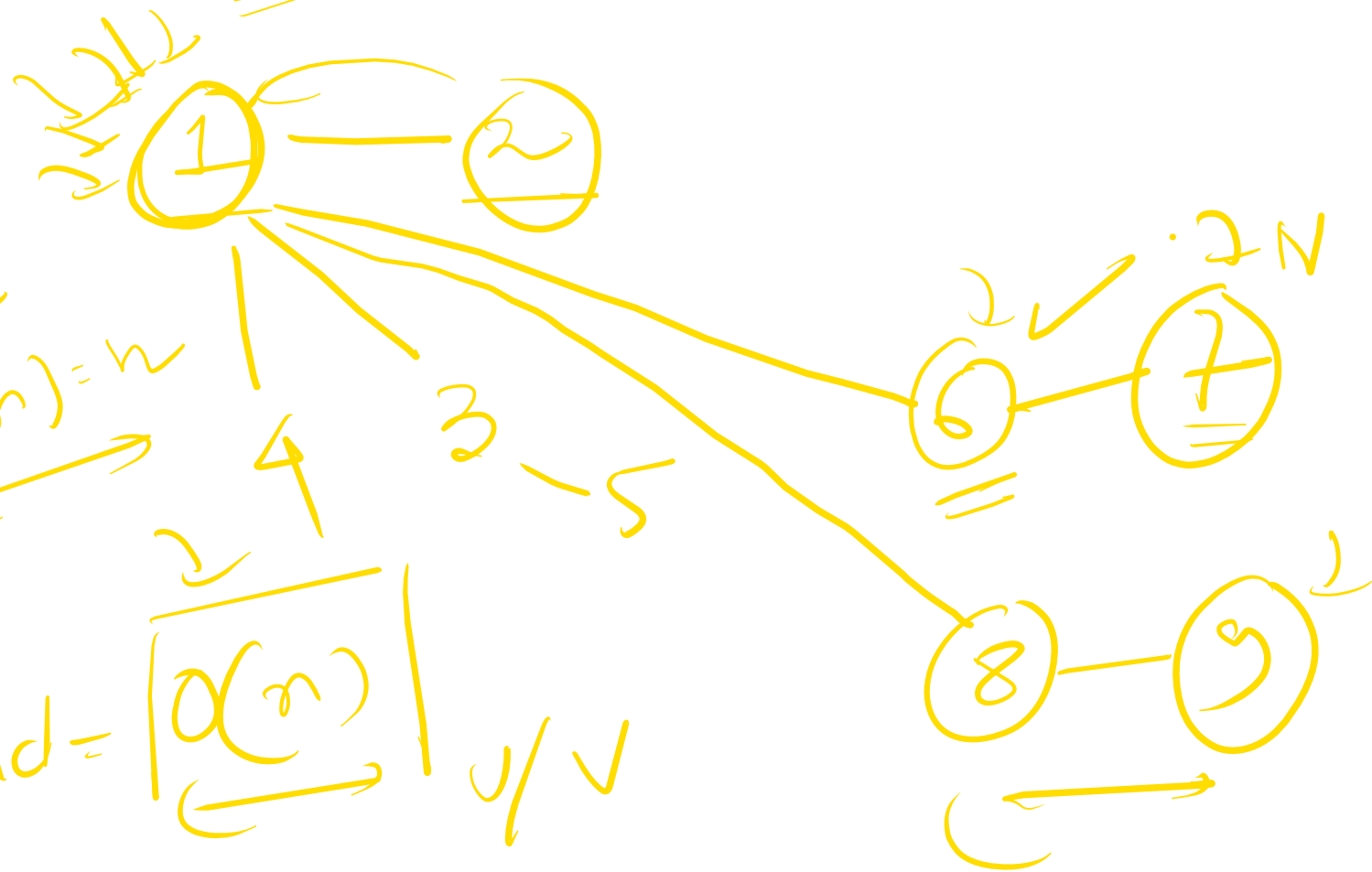
1	2	3	4	5	6	7	8	9
<u>1</u>	2	3	4	5	6	7	8	9
	1	2		3	1	6		8

set

$p \rightarrow$

$n \rightarrow p(r) = n$

$find = \lfloor \frac{O(n)}{2} \rfloor \checkmark \checkmark$



cycle detection condition



↓
cycle is
present?

✓ ✓
1-2

3-4

5-6

7-8

→ 2-4

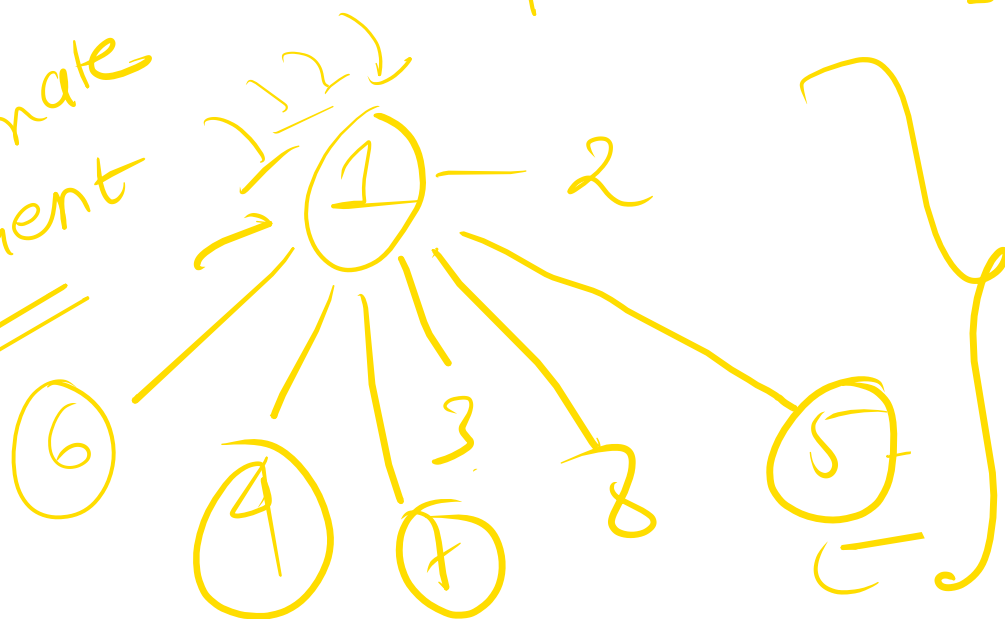
→ 4-6
8-8

Path compression
←

1 2 3 4 5 6 7 8
p: 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8
1 1 3 5 1 1

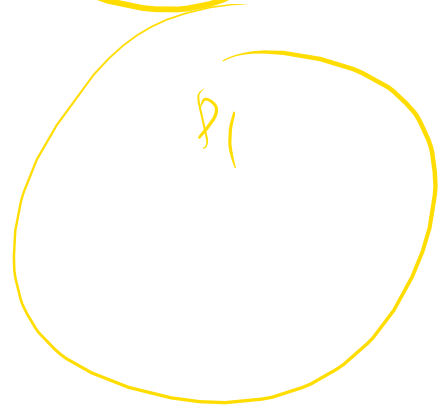
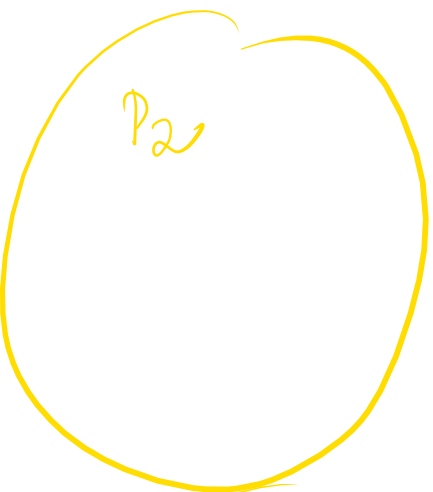
⊕
p[0] = u

ultimate
parent

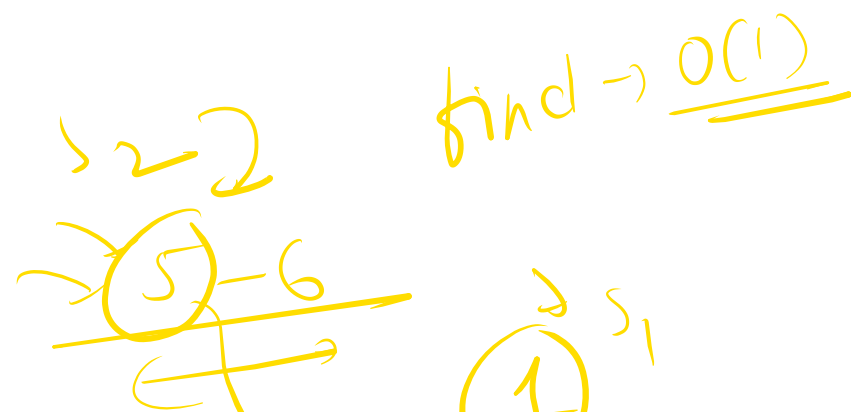


→ flattens w/ the
tree





↓
union by size
union by rank



find → $O(1)$

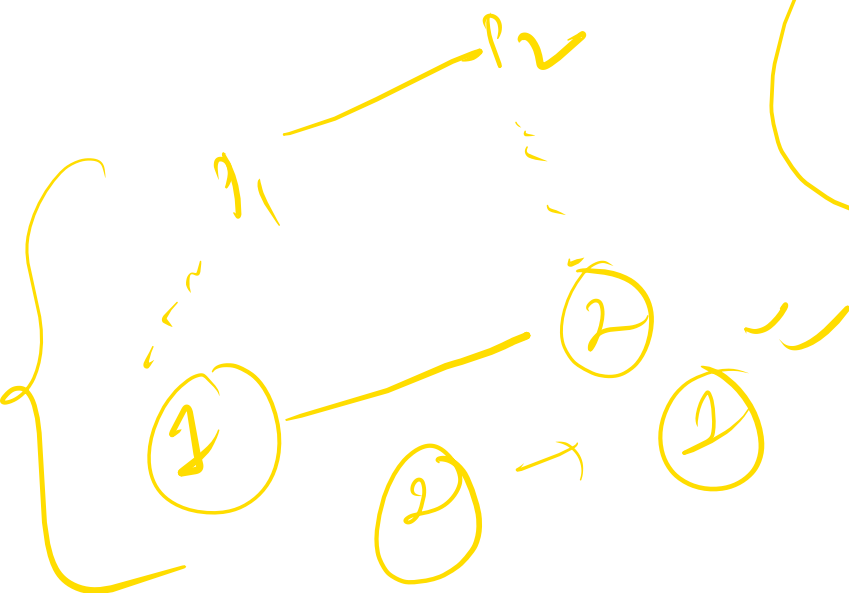


③ → edges ⊕
edges ✓

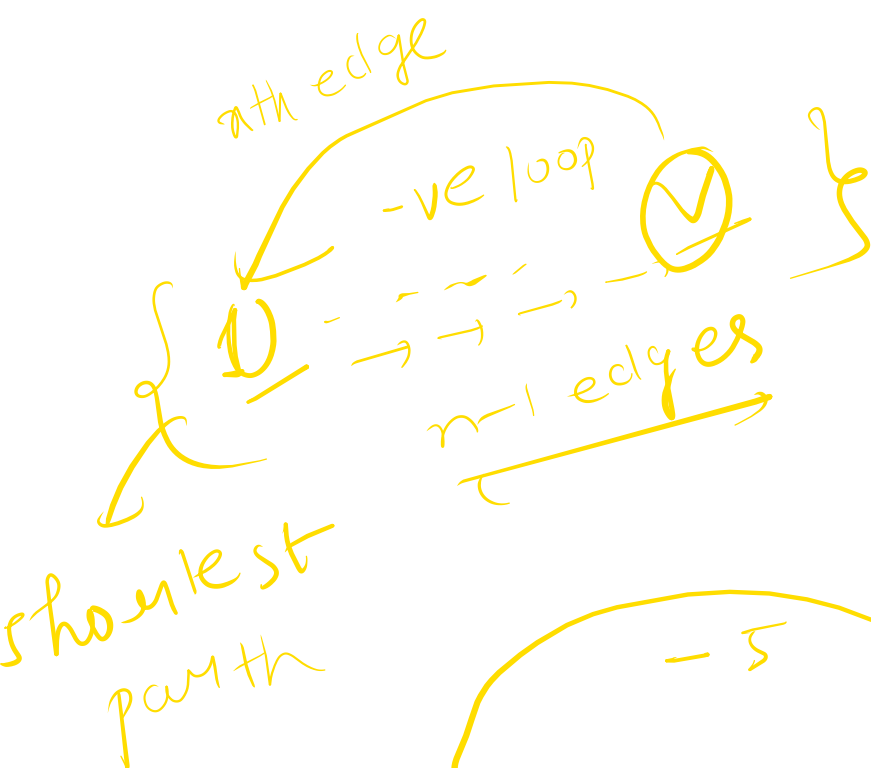
Union based
ops

~~union~~ }
DSU

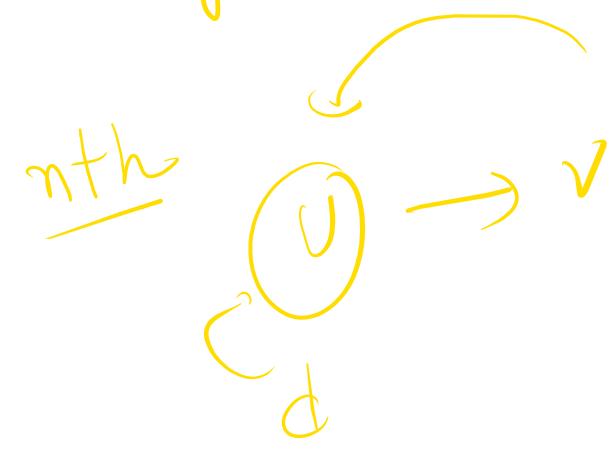
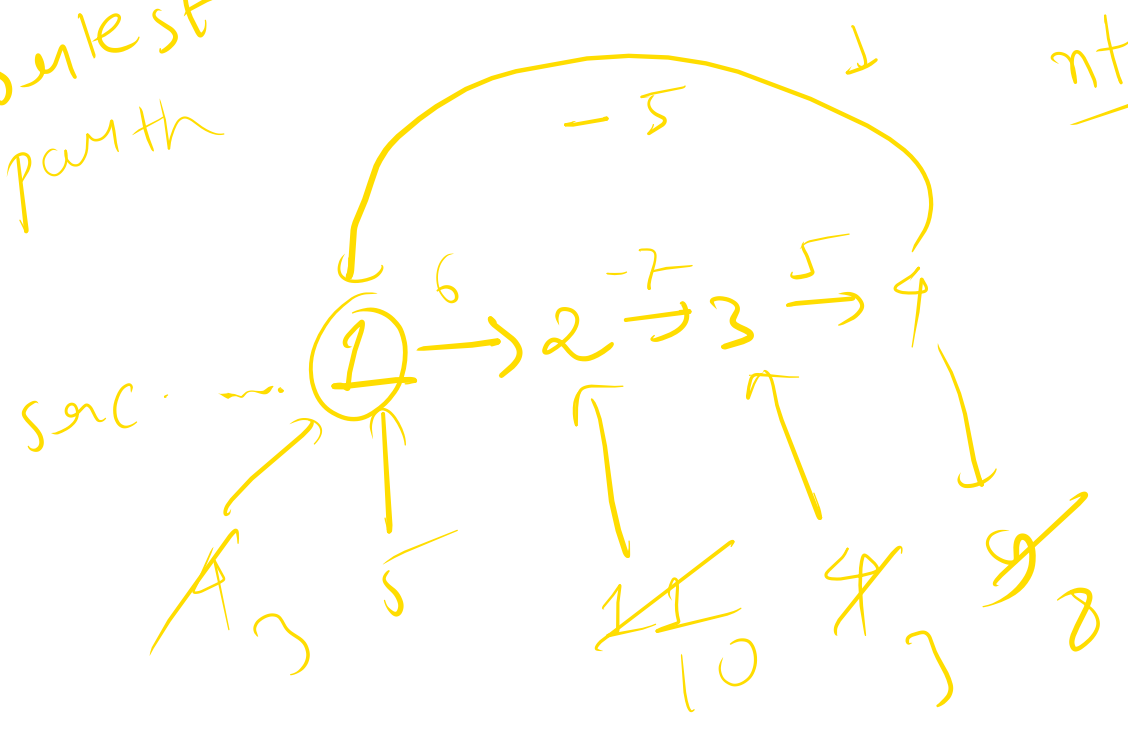
unique.?



+ path compression
+ union by size



Bellman ford
 ↓
 edge relaxation.



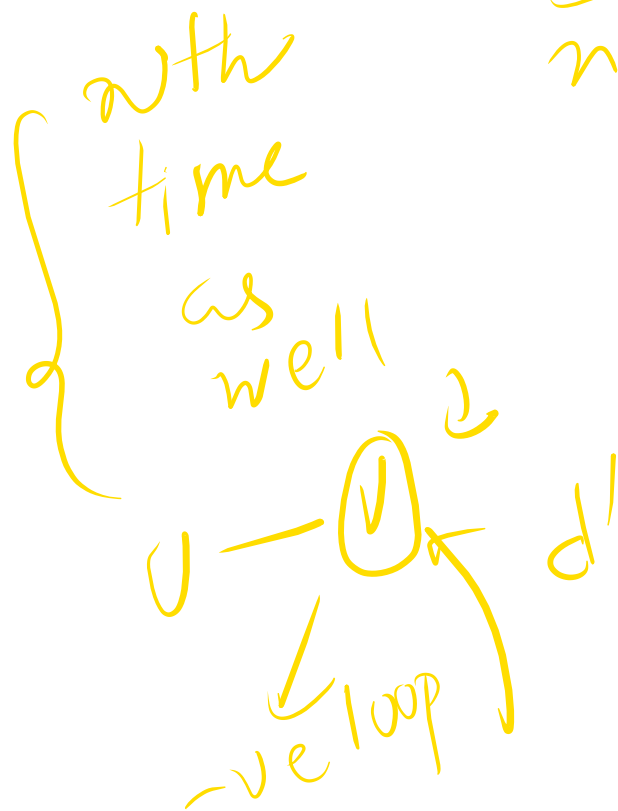


Shortest path

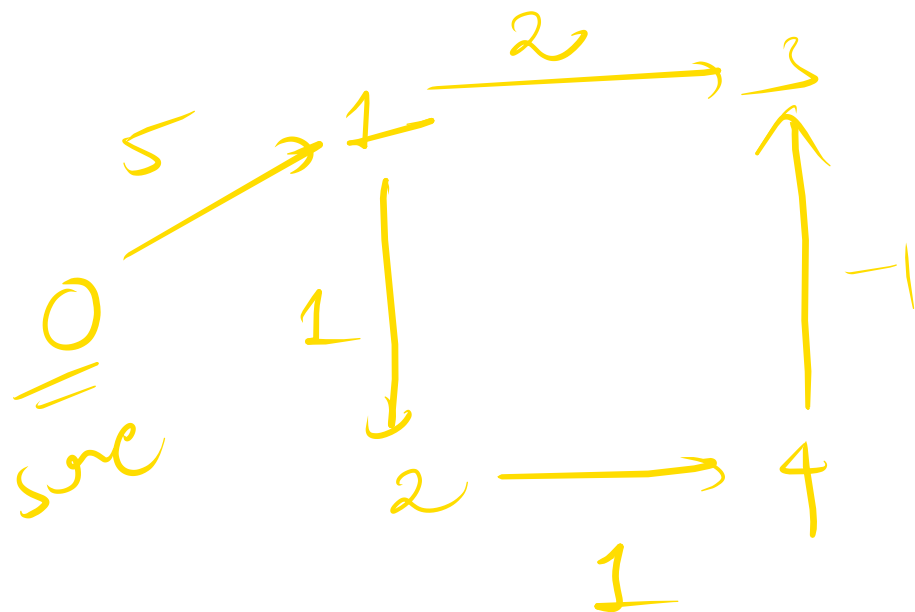
edge
 $\downarrow$
 $n-1$ times.



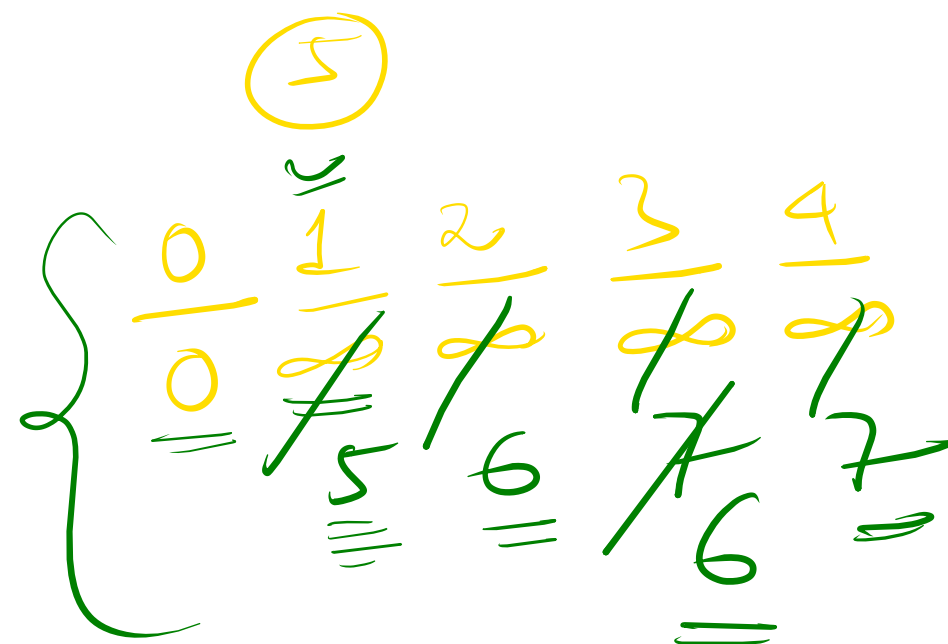
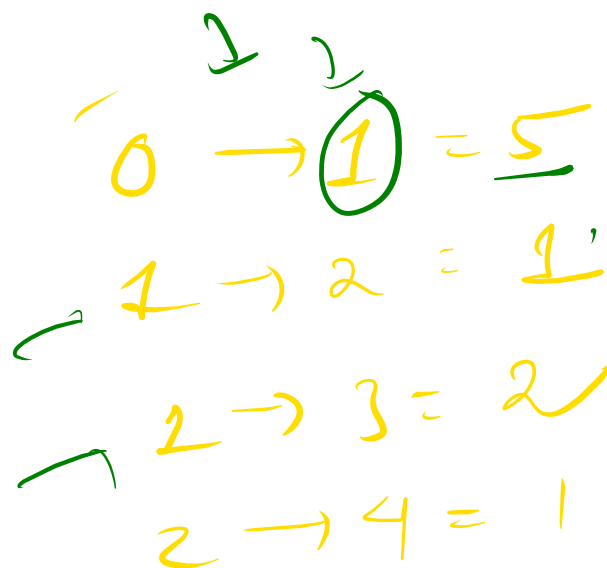
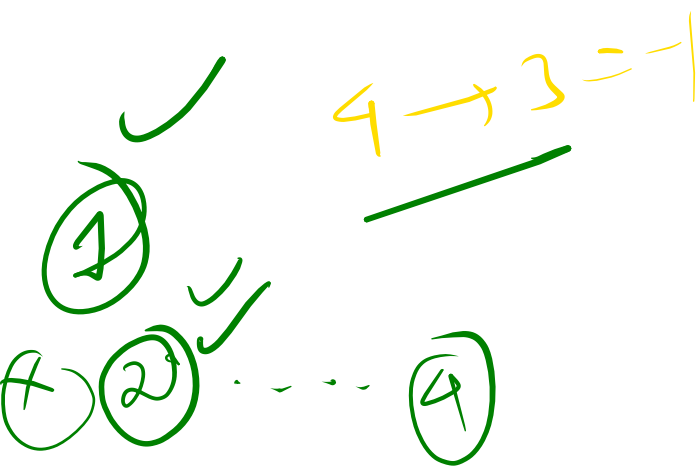
changed



self bond



n-1



n-1 →



lower distance.

